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Powers of One: The Mathematicalization of the Occult Sciences in the High Persianate Tradition*

Matthew Melvin-Koushki

University of South Carolina

mmelvink@sc.edu

Abstract

Occultism remains the largest blind spot in the historiography of Islamicate philosophy-science, a casualty of persistent scholarly positivism, even whiggish triumphalism. Such occultophobia notwithstanding, the present article conducts a survey of the Islamicate encyclopedic tradition from the 4th–11th/10th–17th centuries, with emphasis on Persian classifications of the sciences, to demonstrate the ascent to philosophically mainstream status of various occult sciences (*‘ulūm ḡarība*) throughout the post-Mongol Persianate world. Most significantly, in Persian encyclopedias, but not in Arabic, and beginning with Faḥr al-Dīn Rāzī, certain occult sciences (astrology, lettrism and geomancy) were gradually but definitively shifted from the natural to the mathematical sciences as a means of reasserting their scientific legitimacy in the face of four centuries of anti-occultist polemic, from Ibn Sīnā to Ibn Ḥaldūn; they were simultaneously reclassified as *the* sciences of *walāya*, moreover, which alone explains the massive increase in patronage of professional occultists at the Safavid, Mughal and Ottoman courts in the runup to the Islamic millennium (1592 CE). I argue that the mathematicalization, neopythagoreanization and sanctification of occultism in Ilkhanid-Timurid-Aqqyunlu Iran is the immediate intellectual and sociopolitical context for both the celebrated mathematization of astronomy by the members of the Samarkand Observatory in the 9th/15th century and the resurgence of neoplatonic-neopythagorean philosophy in Safavid Iran in the 10th/16th and 11th/17th, whereby Ibn Sīnā himself was transformed into a neopythagorean-occultist—processes which have heretofore been studied in atomistic isolation.

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Keywords

ʿUlūm ǧarība – *taṣnīf al-ʿulūm* – mathematicalization – mathematization of astronomy – neopythagorean-neoplatonic philosophy – High Persianate – Ilkhanid – Timurid – Aqquyunlu – Safavid – Mughal – Ottoman

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Raising towards heaven our cabbalistic eyes

—JOHN DEE¹

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As their handle implies, the ‘strange sciences’ (*al-ʿulūm al-ǧarība*), that is to say the occult sciences, became a popular site of scholarly controversy from the moment they entered Islamdom in the 2nd/8th century onward.² They long exerted—and still exert—a polarizing effect, inspiring either passionate partisanship or sneering abuse in some of the greatest thinkers and writers of the premodern Islamicate world. On the one hand, anti-occultism polemics have a venerable pedigree in Arabic letters as one of the few subjects where philosophers like Ibn Sīnā, theologians like Ġazālī, traditionalists like Ibn Taymiyya and historians like Ibn Ḥaldūn find common cause. The anti-occultist camp further boasts such luminaries as al-Aṣʿarī, Fārābī, al-Bāqillānī, Qāḍī ʿAbd al-Ġabbār, Birūnī, Ibn Ruṣd, Maimonides, Ibn Qayyim al-Ġawziyya and Ibn al-Šāṭir.³ But the pro-occultist camp is no less replete with intellectual heavyweights; it includes al-Kindī, Abū Maʿšar Balḥī, the Iḥwān al-Šafāʾ, Faḥr al-Dīn Rāzī, al-Būnī, Ibn ʿArabī, al-Maqrīzī, Ibn Turka, ʿAbd al-Raḥmān al-Biṣṭāmī, Mullā Fanārī, Qāzizāda Rūmī, Davānī, Ḥusayn Vāʿiẓ Kāšifī, Mīrim Çelebi, Šams

1 *Monas Hieroglyphica*, trans. Josten, p. 175. The original Latin: “Sublevatis ergo in Cælum oculis Cabalisticis”

2 The standard Arabic term for the occult sciences, including astrology, alchemy and a variety of magical and divinatory techniques, is *ʿulūm ǧarība*, meaning those sciences that are unusual, rare or difficult; less frequently used terms are *ʿulūm ḥafīyya* and *ʿulūm ǧāmīḍa*, sciences that are hidden or occult. More generally, ‘science’ awkwardly translates the Arabic term *ʿilm* (lit., ‘knowledge’), which rather corresponds to the broader concepts of *scientia* and *Wissenschaft*, including *Geisteswissenschaft* (see Williams, “Science”).

3 See Melvin-Koushki, “Defending Geomancy,” n. 65.

al-Dīn Ḥafīrī, Taşköprüzāde, Mīr Faṭḥ Allāh Şīrāzī, Şayḥ Bahā'ī, Mīr Fındıriskī and Mīr Dāmād. Was the duel then a draw?

Hardly: after centuries of pitched debate, occultism's proponents appear to have won, and its opponents to have lost. This denouement took place by the early 9th/15th century, when occult philosophers succeeded in creating a new epistemic style that rendered the occult sciences an essential part of the intellectual, cultural and political mainstream throughout the Islamicate world, largely marginalizing their opposition in the process. The vigorous Arabic anti-occultist tradition here lapses into a curious silence; there are no sustained attacks on the occult sciences after that launched by Ibn Ḥaldūn (d. 808/1406) at the turn of the 9th/15th century.⁴ And even his attack, flailing, fideistic and overlong, conveys an air of desperation. Tellingly, it failed to convince his student and sole defender, al-Maqrīzī (d. 845/1442), who was an enthusiastic occultist.⁵ More sophisticated and philosophically convincing are Ibn Qayyim al-Ğawziyya's (d. 751/1350) sorties of several decades earlier,⁶ but here too the Hanbali jurist-theologian comes off as embattled and reactionary, and his categorical association of neopythagorean-neoplatonic (*fiṭāğūrī-aflātūnī*) occultism with Avicennan philosophy as the two greatest threats to Islam in the post-Mongol era simply testifies to the ascendancy and interpenetration of both in his own day. (He presents as model of the modern reprobate intellectual Naşīr al-Dīn Ṭūsī (d. 672/1274), the great destroyer of Islam, a philosopher-astronomer who heavily patronized astrologers, naturalists and mages and ended his days a sorcerer.⁷) The loud yet anxious anti-occultist platforms of both men should thus be taken not as evidence for occultism's marginalization, as is usually done,⁸ but rather as heralding an occultist renaissance.

Very like contemporary and later developments in Renaissance Europe, this florescence of occultism was equal parts philosophical and political in stem-

4 His anti-occultism arguments, constituting a full 11 percent of the *Muqaddima* as a whole, occur primarily in the sixth premise (*muqaddama*) of chapter one and chapter six on the sciences. This is not to suggest that there were not episodic attacks on certain occult sciences by jurists thereafter, but these tended to be brief and confined to the Ottoman context; see below.

5 In his *Muqaffā*, for instance, al-Maqrīzī goes so far as to valorize Ḥasan-i Şabbāḥ (d. 518/1124) himself, attributing his political genius to his mastery of various occult sciences (*ğawāmiḍ 'ulūm*) (vol. 3, pp. 327–334, no. 1160; my thanks to Carol Hillenbrand for this reference).

6 Ibn al-Qayyim's attacks on occultism are a prominent feature of both *Miftāḥ dār al-sa'āda* and *Iğāṭat al-laḥfān min maşāyid al-şayṭān*; see Livingston, "Science and the Occult."

7 *Iğāṭat al-laḥfān*, pp. 260–266.

8 See e.g. Dallal, *Islam*, pp. 144–145. For an examination of the anti-occultism arguments of Ibn al-Qayyim and Ibn Ḥaldūn see Melvin-Koushki, "Defending Geomancy."

ming. In the first place, the Mongol conquest of western Asia in the mid-7th/13th century destroyed the caliphal-sultanic-jurisprudential model that had become standard in Islamicate societies, though seriously compromised by the time of the Mongols' arrival; into the sociopolitical vacuum thus created rushed the category of sainthood or sacral power (*walāya*), particularly in the guise of sufism on the one hand and occultism on the other, which rapidly attained hegemonic status throughout the Turko-Mongol Perso-Islamic world in particular (i.e., Anatolia to Central Asia and India).⁹ From the 8th/14th to the 11th/17th century messianic sultans and saints competed fiercely with one another in laying claim to this sacral power: sultans became saint-kings (as in the Mughal and Ottoman Empires), and sufis became sultans (as in the Safavid realm). It is here significant that Ibn 'Arabī (d. 638/1240), the hugely influential 7th/13th-century Andalusian philosopher-mystic and foremost theoretician of *walāya*, was recruited as symbolic linchpin of High Persianate¹⁰ imperial ambitions, especially by the Ottomans; Yavuz Selīm (r. 918–926/1512–1520) declared Ibn 'Arabī the patron saint of the Ottoman Empire upon his conquest of Damascus in 922/1516. Sacral power aside, the (in)famous Ibn 'Arabian doctrine of the oneness of being (*waḥdat al-wuḡūd*) was likewise highly adaptable to expansive and universalizing imperialist ideologies.¹¹

This on the eastern front; different but complementary processes were afoot on the western front. Most notably, the 7th/13th century marks the beginning of an unprecedented eastward flow of scholars and mystics from the Maghrib in the face of the Reconquista to the north and the rise of the puritanical Almohad regime (524–668/1130–1269) to the south.¹² The Mongol invasion, of

9 Mir-Kasimov, "Conflicting Synergy of Patterns," pp. 11–13.

10 I use this term to denote the Persian cosmopolis during the 8th–14th/14th–20th centuries, which featured a high degree of cultural and intellectual continuity, consciously cultivated by mobile, autonomous scholarly elites, that was profoundly, if unevenly, disrupted by European and Soviet colonialism. While this period generally corresponds to European "early modernity" and "modernity," I avoid these terms here as too eurocentric and teleological for use in the Persianate context.

11 This doctrine was also invoked in challenge to imperialist ideologies, this most famously by Shaykh Badr al-Dīn of Simavna (executed 819/1416), the *ḥalīfa* of Sayyid Ḥusayn Aḥlāṭī in Anatolia; this jurist-occultist inspired the most successful rebellion in Ottoman history with his platform of radical Ibn 'Arabian monism as developed in the *Wāridāt*, to be expressed religiopolitically and economically in the form of pure communism and monotheistic communion among the impoverished Muslim and Christian peasantry of Rumelia and armed resistance to oppressive Ottoman elites.

12 It should be noted, however, book burnings aside, that the Almohad reputation for puritanical fanaticism is likely a polemical exaggeration on the part of discomfited Andalusī

course, was simultaneously driving many of their persophone counterparts to flee westward. As a consequence, in the 7th/13th and 8th/14th centuries Ayyubid-Mamluk Egypt and Syria became the locus of a highly creative mixing of eastern and western intellectual and mystical traditions that had developed semi-independently for centuries.¹³ The North African divinatory science of geomancy (*ʿilm al-raml*), for instance, Arabic answer to the *I Ching*, is unattested in Iran before the late 6th/12th century, but there burgeoned in popularity from the Ilkhanid period onward.¹⁴ Ibn ʿArabī himself is the poster child for this new ferment: an Andalusian refugee in Damascus whose most important disciples were Persian, such that the Ibn ʿArabian tradition sunk far deeper roots in the Persianate east than the Arabicate west. During this period, Mamluk Cairo emerged as one of the most important intellectual hubs of the Islami-cate world, attracting scholars from far and wide, particularly Anatolia and Iran; in this capacity it became the epicenter of the occultist renaissance here under investigation.¹⁵ The crucial factor in this connection was the secretive study circles or esotericist reading communities that coalesced around the writings of the Ifriqian sufi Aḥmad al-Būnī (d. 622/1225?), the greatest mage of Islam; he is best known as the putative author of the *Great Sun of Knowledge* (*Šams al-maʿārif al-kubrā*), by far the most seminal grimoire in the Islamicate world down to the present.¹⁶ Following the general 7th/13th-century trend (as well as

princes, an account tendentiously seized upon and further amplified by 19th-century historians (my thanks to Nicholas Harris for this observation).

13 Muhanna, "Encyclopædism," pp. 44–45.

14 See Melvin-Koushki, "Persianate Geomancy."

15 On Cairo's new status as the "political and cultural epicenter of the Muslim world" see Muhanna, "Encyclopædism," pp. 44–45; Petry, *Civilian Elite*; Berkey, *Transmission*, ch. 1.

16 As Noah Gardiner has established, the *Šams al-maʿārif al-kubrā*, to be distinguished from al-Būnī's authentic *Šams al-maʿārif wa-laṭāʾif al-ʿawārif*, is largely a pseudograph product of the 10th/16th century at the earliest, and to be partially associated with 9th/15th-century Ottoman occultists like ʿAbd al-Raḥmān al-Bisṭāmī ("Esotericism in a Manuscript Culture," pp. 33–37). It is more immediately utilizable than al-Būnī's authentic works, which are heavily sufi in orientation, inasmuch as it represents a later, more mature phase of sufi magic. In its immense popularity, the *Šams al-maʿārif al-kubrā* functioned—and continues to function—in the Islamicate world as the *Picatrix* did in the Christianate; ironically, the latter grimoire's Arabic original, the *Gāyat al-ḥakīm* of Maslama al-Qurṭubī (d. 353/964), was far less seminal in arabophone (and persophone) regions, presumably due to the relative geographical isolation of al-Andalus vis-à-vis the Islamicate heartlands during this period. At the same time, it seems to have been rediscovered in Safavid Iran—it is often cited by thinkers like Mir Dāmād, Mullā Ṣadrā and Qāzī Ṣāʿid Qummī to corroborate basic philosophical ideas (my thanks to Sajjad Rizvi for this observation).

the itinerary of al-Būnī himself), these circles appear to have migrated eastward from Ifriqiya and established themselves in Cairo by the end of that century. The manuscript record suggests that the study of al-Būnī's works remained a modest, secretive affair until the mid-8th/14th century, at which juncture and for as yet unknown reasons interest exploded among Mamluk elites, who began to vigorously patronize the production and audition of Būnian texts;¹⁷ this mainstreaming of sufi magic was then recorded by Mamluk encyclopedists like al-Nuwayrī (d. 733/1333) and al-Qalqašandī (d. 821/1418), whose open and decontextualized quotation of passages from al-Būnī's works reflects an unprecedented *de-esotericization* and *adabization* of occultism.¹⁸ This development drew professional occultists to the capital, including most notably the Tabrizi Kurdish *magus* Sayyid Ḥusayn Aḥlāṭī (d. 799/1397), personal physician-alchemist to al-Malik al-Ẓāhir Barqūq (r. 784–801/1382–1399), conscious heir of both al-Būnī and Ibn 'Arabī and master to the most important occult philosophers of the early 9th/15th century.¹⁹ It is precisely this explosion of interest in occultism at the highest echelons of Mamluk society that so discomfited both Ibn al-Qayyim and Ibn Ḥaldūn and inspired their worried diatribes.²⁰

- 17 Gardiner, "Esotericism," pp. 145–160. The Black Death epidemics, which began recurring in Egypt from the mid-8th/14th century onward, created an apocalyptic mood at all levels of society and inaugurating a chaotic period of social leveling, and hence were almost certainly a significant factor in this connection.
- 18 Gardiner, "Esotericism," pp. 281–293. Al-Nuwayrī clearly conceives of lettrism as a natural science, though his classification scheme is strictly implicit. In the *Nihāyat al-arab*, at the end of the fourth book on plants, the three subsections of the eleventh and final subchapter of its fifth chapter cover the activation of occult properties of substances (*ḥawāṣṣ*). Its first subsection covers matters pertaining to women, sex and marriage; the second talismans; and the third lettrism (*ḥawāṣṣ al-ḥurūf wa-l-asmā'*), consisting almost entirely of a series of excerpts from al-Būnī's *Laṭā'if al-iṣārāt*. Similarly, al-Qalqašandī classifies under the rubric of the natural sciences magic (*'ilm al-siḥr*), lettrism and magic squares (*'ilm al-ḥarf wa-l-awfāq*), talismans, physiognomy, oneiromancy, judicial astrology (*aḥkām al-nuġūm*) and geomancy (*'ilm ḍarb al-raml*), the last being the speciality of bedouins (*al-'arab*) (Gardiner, "Esotericism," pp. 283, 292).
- 19 Gardiner, "Esotericism," pp. 75–77. On Barqūq's court as hotbed of occultist practice, with Aḥlāṭī as ringleader, see Cornell Fleischer's forthcoming translation of 'Abd al-Raḥmān al-Biṣṭāmī's *Durrat al-āfāq fi 'ilm al-ḥurūf wa-l-awfāq*.
- 20 Ayyubid patronage of occultism seems not to have been insignificant—Šalāḥ al-Dīn (r. 564–589/1169–1193), for instance, is often reported in chronicles and biographical dictionaries as being interested in alchemy as a scientific pursuit, with Suhrawardī himself being said to have gained entrance at his court by showing alchemically produced gems (personal communication with Sonja Brentjes and Nicholas Harris)—, but the subject has yet to be investigated. Ayyubid-Mamluk Egypt and Syria aside, Ibn al-Qayyim was equally con-

On the philosophical front, the same fraught but fecund post-Mongol period saw a shift away from Avicennan peripateticism and toward neoplatonism and neopythagoreanism, the natural bases for all high occultist theory and practice.²¹ This epistemic shift represented, in the first place, a conscious recuperation of the overtly neopythagorean-occultist project of the 4th/10th-century Brethren of Purity, who in their *Epistles* designate occult sciences like astrology, alchemy and magic the apex of intellectual-spiritual ascent and ultimate goal of the illumined philosopher-saint. It also entailed a new embrace of Ibn ‘Arabī as the most powerful neopythagorean-occultist thinker since the Brethren. Not incidentally, such a move had the benefit of jibing far better with the newly hegemonic category of *walāya*. The ascendancy of both the Ibn ‘Arabian and the Būnian traditions, the first theoretical and the second practical,²² thus represents the full realization of the Brethren’s project: the *sanctification* of occultism, until this point merely a subset of aristotelian natural philosophy.²³ At the same time, centuries of criticism from more traditionalist theologians prompted occult and non-occult scientists alike to take stock of their disciplines and develop new, synthetic methodologies that could more deftly sidestep such critiques.²⁴

Among the most important scientific-philosophical outcomes of this new neopythagorean sensibility is the *mathematical turn* pioneered by a series of prominent astronomer-cosmologists. Beginning in the early 9th/15th century, the astronomers and mathematicians associated with the famed Samarkand Observatory of Uluġ Beg (r. 811–853/1409–1449), including in the first place Qāzizāda Rūmī (d. 835/1432) and his student ‘Alī Qūščī (d. 878/1474), sought to conclusively rid astronomy of its longstanding dependency on aristotelian physics and reformulate it in purely mathematical terms—the freeing of astronomy from philosophy, as Jamil Ragep has summarized their project.²⁵ In so doing, they were consciously continuing the mathematizing project of ear-

cerned about the rising occultist tide in the post-Mongol east, about which Ibn Ḥaldūn is far more poorly informed.

- 21 I here define neopythagoreanism generically as the explicit invocation of Pythagoras and his intellectual heirs, as well as an adherence to the doctrine—reliably associated with Pythagoras by High Persianate occultists—of number-letter as the uncreated, all-creative matrix of the cosmos and the intellect’s only path of return to the One.
- 22 Gardiner, “Esotericism,” p. 265; Melvin-Koushki, “The Quest,” p. 164.
- 23 Liana Saif shows the mechanics of this process in a forthcoming article, “From *Ġāyat al-ḥakīm* to *Šams al-ma’ārif*.”
- 24 See Ragep, “Freeing Astronomy.”
- 25 Ragep, “Freeing Astronomy.”

lier astronomers, especially Abū Rayḥān Bīrūnī (d. c. 442/1050) and Quṭb al-Dīn Šīrāzī (d. 710/1311), who also asserted the epistemological superiority and independence of their discipline vis-à-vis natural philosophy and metaphysics—as did Ptolemy himself; yet these latter did not go so far as to simply ignore aristotelian inputs altogether in the manner of their Timurid heirs. Most impressively, this mathematical turn is emblemized by Qūščī's status as the first astronomer in the Islamo-Christian tradition to seriously defend the theoretical possibility of a rotating earth, anathema to aristotelian physicists, thereby serving (along with Aristarchus (d. c. 230 BCE)) as crucial precedent for Copernicus.²⁶ This fully mathematized astronomy continued to be pursued with vigor throughout the Persianate world into the 10th/16th century, which hosted such philosopher-scientists as Mīr Faṭḥ Allāh Šīrāzī (d. 997/1589), polymath extraordinaire to Emperor Akbar (r. 963–1014/1556–1605) in India; Šayḥ Bahā'ī (d. 1030/1621), a founder of the philosophical school of Isfahan and charismatic architect of the new Safavid Shī'i imperial culture; and Šams al-Dīn Ḥafīrī (d. 942/1535), identified by George Saliba as the most important planetary theorist of that century and the developer of a new mathematical language for analyzing the cosmos that seems precociously “modern.”²⁷ Building on Ragep's argument in particular, İhsan Fazlıoğlu posits this trend as an unprecedented form of “mathematical humanism” predicated on the principle that mathematical models are not simply conjectural or imaginary constructs, as had been held from Ibn Sīnā onward, but can in fact correspond to reality; and with certainty in mathematical knowledge depending solely on the degree of mathematical exactitude achievable in a given model, it was therefore now scientifically possible and indeed preferable to bypass both metaphysics and aristotelian natural philosophy in practicing cosmology.²⁸

Similarly important was the thoroughgoing *neoplatonization* and *neopythagoreanization* of philosophy itself in Safavid Iran, by which Pythagoras, Plato and Plotinus were combined with Ibn 'Arabī to explode the strictures of Avicennan and illuminationist metaphysics and create a new, transcendent philosophy (*ḥikma muta'āliya*) that was not merely Greek but simultaneously perennial and Twelver Shī'i. Mīr Dāmād (d. 1040/1630), the Third Teacher, is usually identified as the ringleader of this effort; its greatest architect was his student

26 Ragep, “Copernicus,” pp. 138 f; Morrison, “A Scholarly Intermediary.”

27 Saliba, “A Sixteenth-Century Arabic Critique”; idem, “Kāfī, Šams-al-Dīn,” *EIr*. For a synthetic overview of Ragep's and Saliba's arguments that rather emphasizes the scientific continuity obtaining from Bīrūnī and Quṭb al-Dīn Šīrāzī to Qūščī and Ḥafīrī, see Dallal, *Islam*, pp. 58–89.

28 “Between Reality and Mentality,” pp. 35–37.

Mullā Ṣadrā Ṣhīrāzī (d. c. 1045/1636), the High Persianate answer to Ibn Sīnā. Indeed, the definitive triumph of this new philosophy is best seen in the fact that Ibn Sīnā himself, second father of peripateticism, was transformed into a neopythagorean during the 10th–11th/16th–17th centuries;²⁹ so too was Aristotle, this through a great surge in popularity of such pseudepigraphical texts as the *Uṭlūḡīyā* or *Theology of Aristotle*, a 2nd/9th-century Arabic paraphrase of Plotinus' *Enneads* IV–VI, which Mīr Dāmād declares to be fundamental to his entire philosophical system.³⁰

These developments are well-known and have been suitably feted in the literature.³¹ What has largely escaped scholarly notice to date, however, is the fact that this neopythagorean-mathematical turn was intimately connected with, even partially driven by, the occultist renaissance of the 8th/14th and 9th/15th centuries. That is to say, all the pioneering astronomer-mathematicians and neopythagoreanizing philosophers named above were also committed occultists or closely associated with the leading occult philosophers and practitioners of their day. Qāẓizāda Rūmī, for example, was a friend and colleague to Ibn Turka (d. 835/1432), chief judge of Isfahan and Yazd and the most impor-

29 (Ps.-)Ibn Sīnā's *R. al-Nayrūzīyya* or *New Year Treatise* represents a brief recapitulation of the philosopher's system in neopythagorean-lettrist terms; the text's authenticity aside, which is not out of the question, it exploded in popularity among Safavid elites by reason of thematic fit. The *R. al-Nayrūzīyya* seems to have been largely ignored in lettrist circles until the 10th/16th century, at which point Ġīyāṣ al-Dīn Daštakī quoted an earlier summary of it in his *Maṭla' al-'irfān fī tafsīr al-Qur'ān*; but Mīr Dāmād was its greatest champion, even to the extent of declaring the work his primary model in his *Ġazāvāt u mavāqīt*, and fully transcribing its second and third sections in the same work (pp. 102, 162, 216, 221, 247, 260, 266). That it was received as an essentially lettrist text is suggested by the fact that in at least one anthology, MS Majlis 10319, it is copied together with three of Ibn Turka's lettrist works; see Melvin-Koushki, "The Quest," pp. 204–209. Ibn Sīnā was also transformed into an occultist after the fact through the attribution to him of the *Kunūz al-mu'azzimūn*, a seminal manual of celestial and terrestrial magic, on which see Ġalāl al-Dīn Humā'ī's introduction to his edition of the text (Tehran: Anḡuman-i Āṣār-i Millī, 1331 š./1952). Taṣkōprūzāde (on whom see below) cites him as a noted authority on *sīmīyā*, together with Suhravardī (*Miftāḥ*, vol. 1, p. 317). Even Ġazālī appears to have been refashioned as an occultist, and particularly a lettrist, by the Safavid period, putative author of texts like the *K. al-Sirr al-maṣūn wa-l-ġawhar al-maknūn*, on the comprehensive prognosticon of 'Alī (Savage-Smith, "Magic and Islam," vol. 1, p. 66, cat. 21, no. 4; Melvin-Koushki, "The Quest," p. 246 n. 279, p. 274).

30 *Ġazāvāt u mavāqīt*, p. 15.

31 See e.g. Rizvi, "(Neo)Platonism"; Adamson, *The Arabic Plotinus*. On Ps.-Aristotle as an important source for Christianate occultism see Thorndike, *History*, vol. 2, pp. 246–278.

tant occult philosopher of early 9th/15th-century Iran.³² Trained by Aḥlāṭī in Cairo and preeminent exponent of the Ibn ‘Arabian tradition, Ibn Turka has the distinction of producing the first *summa* of Islamic neopythagoreanism, the *Book of Inquiries* (*K. al-Mafāḥiṣ*), completed in 823/1420—that is to say, the earliest manifesto for the epistemic shift just outlined. The Isfahani thinker’s primary focus therein is the kabbalistic science of letters (*‘ilm al-ḥurūf*), or lettrism, which he advances as both the science of *walāya par excellence* and the only valid vehicle for metaphysics, thus trumping both sufism and peripatetic-illuminationist philosophy. His student, friend and fellow occultist Šaraf al-Dīn Yazdī (d. 858/1454), best known as dynastic historian to the Timurids, was (naturally) also an accomplished mathematician, and even worked for a time at the Samarkand Observatory.³³ It is precisely this context in which the “geometrical theology” of Qāzizāda—often called, not emptily, the Plato of his age (*Aflātūn al-zamān*)—is to be understood.³⁴ Later in the same century, the prominent philosopher Ġalāl al-Dīn Davānī (d. 908/1502), ideological mainstay to the Aqquyunlu Empire, vaunted Ibn Turkian-Ibn ‘Arabian lettrism as the highest philosophical science and developed a lettrist-astrological platform in support

32 See Melvin-Koushki, “The Quest,” pp. 41, 50 f., 111, 146 f., 225, 441, 569 f.; Eichner, “Afḍal al-Dīn Turka,” *ET*³; Ragep, “Qāḍī-zāda Rūmī,” *ET*². Ibn Turka and Qāzizāda seem to have been friends from childhood, having studied together in Samarkand under the former’s older brother Šadr al-Dīn, an eminent scholar of hadith, tafsir, jurisprudence and *uṣūl al-dīn*, during the first phase of their education. More significantly, Qāzizāda, like Ibn Turka, is reported to have been a disciple to Aḥlāṭī in Cairo (Kamāl al-Dīn Ḥusayn Ġazurgāhī (fl. 909/1504), *Mağālis al-‘uṣṣāq*, p. 189; ‘Abd al-Razzāq Kirmānī (fl. 900/1495), *Tazkira dar Manāqib-i ḥaẓrat-i Šāh Ni‘mat Allāh Valī*, pp. 70 f.). In 829/1426 Ibn Turka dedicated his lettrist work *Šarḥ al-Basmala* to Uluğ Beg and sent a copy to Qāzizāda Rūmī, suggesting his friend’s continued interest in matters lettrist; the strength of their association is confirmed by the astronomer’s warm letter of thanks to Ibn Turka for sending this work (transcribed in Melvin-Koushki, “The Quest,” pp. 569 f.). Qāzizāda opens this letter with tones of great respect, including descriptive phrases emphasizing Ibn Turka’s mastery of the whole of the Islamic tradition (*ḥaẓrat-i šarī‘at-šī‘ārī-yi ṭarīqat-dišārī-yi ḥaqīqat-āšārī*) and spiritual perfection (*valāyat-šī‘ārī-yi virāṣat-dišārī-yi ḥatmī-yi atammī*), and expresses his great desire to see him—the alchemical matrix of everlasting life and eternal collyrium (*kīmīyā-yi ḥayāt-i abadī u tūtīyā-yi binā‘ī-yi sarmadī*)—in person; he then mentions meeting Ibn Turka in Herat and reminds him of his promise to visit him in Samarkand. In brief, the two thinkers were hardly passing acquaintances, but rather close associates during their lifetimes and remembered as such until at least the early Safavid period.

33 Binbaş, *Intellectual Networks*, p. 65; Monfared, “Sharaf al-Dīn ‘Alī Yazdī.”

34 Fazlıoğlu, “The Samarqand Mathematical-Astronomical School,” p. 27.

of his patrons' political legitimacy.³⁵ Likewise, the astronomer-mathematician and Safavid *shayḥ al-islām* Šayḥ Bahā'ī is remembered to this day as the most powerful practicing occultist of the 10th/16th century, having learned the talismanic arts from Maḥmūd Dihdār Širāzī (fl. 676/1569), aspiring courtier to Shah Ismā'īl II (r. 984–985/1576–1577) and the most prolific Persian writer on lettrism of the same century.³⁶ And like most of his colleagues, Ḥafīrī, mathematical revolutionary and planetary theorist extraordinaire—as well as, not incidentally, Maḥmūd Dihdār's father—, penned treatises on both geomancy and lettrism as a matter of course.³⁷

In sum, not only did our pioneering astronomer-mathematicians and neopythagoreanizing philosophers not consider the occult sciences peripheral or antithetical to their respective projects, they argued for their indispensableness as logical extensions and practical applications of neopythagorean-mathematical theory. This intellectual turn also happened to smoothly accord with the imperial ideologies developed by Ottoman, Safavid and Mughal dynasts, many of whom heavily patronized these occult philosophers and occult scientists as props to their political legitimacy and allies in the contest for sacral power. In other words, the *ġarīb* sciences were now considered not strange but fundamentally *elite*.³⁸ By amputating occultism from our narratives of Islamicate intellectual and political history, then, or simply quarantining science and statecraft from sainthood, we run the risk of fundamentally misunderstanding High Persianate culture as a whole.

The Mathematicalization of the Occult Sciences

Unsurprisingly, the shift away from Avicennan peripateticism and toward a mathematized philosophy-science can be tracked in the changing epistemological status of the occult sciences in Arabic and Persian classifications of the sciences (sg. *taṣnīf al-ʿulūm*). The overall arc represents a gradual deemphasis of the aristotelian model established by Fārābī (d. 339/950) and Ibn Sīnā (d. 428/1037), wherein the occult sciences are classified exclusively as natural sciences, and an increasing preference for a more neopythagorean model

35 See Melvin-Koushki, *The Occult Science*; Woods, *Aqquyunlu*, pp. 102–104, 145.

36 See Melvin-Koushki, “Maḥmūd Dehdār Širāzī,” *Elr*; idem, *The Occult Science*.

37 Saatchian, *Gottes Wesen*, pp. 51–54.

38 Cf. Gāzurgāhī's reference to Ahlāṭī as consummate occultist: “This holy man possessed total mastery of the recondite points of elite sciences (*ġavāmīz-i ʿulūm-i ḥāṣṣ*)” (*Maḡālīs al-ʿuṣṣāq*, p. 189).

that privileges the mathematical sciences and reclassifies many of the occult sciences under that rubric—a process I here term *mathematicalization*. Not coincidentally, this process climaxed in the Persianate east with the mathematical turn of the 9th–10th/15th–16th centuries. It was less a linear development, however, than a cyclical one: the ascent from essentialist aristotelianism to existentialist neoplatonism-neopythagoreanism was also a return to the same, that is, as established by the Iḥwān al-Ṣafā' already in the 4th/10th century. Yet the Avicennan model remained hugely influential, its epistemological hierarchy—logic, physics, mathematics, metaphysics—readily adopted even by later occultists.³⁹ That is to say, the complex reception history of both aristotelian and neoplatonic-neopythagorean currents in the Islamicate world entailed their intermixture to the point of inextricability; even Ibn Sīnā, that second Aristotle, was a neoplatonist on occasion—as was Aristotle himself, this in the guise of seminal pseudepigrapha like the *Uṭūlūḡiyyā*, *K. al-Tuffāha* (*Liber de pomo*), *Kalām fī maḥd al-ḥayr* (*Liber de causis*), *Asrār al-nuḡūm* (on astrology) and *Sirr al-asrār* (*Secretum secretorum*), all of which burgeoned in popularity in Safavid Iran in particular.⁴⁰ The meteoric rise of sufism to hegemonic status in the post-Mongol period helped to tip the balance in favor of neoplatonism, to be sure (sufi theory after Ibn 'Arabī being thoroughlygoingly neoplatonic by default). But the choice to emphasize one or the other remained very much a matter of personal preference, with the ultimate goal of many philosophers, from Fārābī onward, being to reconcile Plato and Aristotle.⁴¹

Given these two philosophical poles, a perennial tension obtained between the natural and mathematical sciences in classifications of the sciences, particularly in the Persianate world. İhsan Fazlhoğlu summarizes the epistemological stakes as follows, here as context for the construction of the Samarkand Observatory:

According to ancient theories, the language of mathematics belonged to the celestial region, which was made up of aether, moved with circular

39 Muhanna, "Encyclopaedias, Arabic," *ET*³.

40 Pourjavady and Schmidtke, "An Eastern Renaissance?" pp. 258–260. On Aristotle as a primary authority on astrological magic see Burnett, "Arabic."

41 The Ps.-Fārābī text *al-Ġam' bayn ra'yay al-ḥakīmayn Aflāṭūn al-ilāhī wa-Aristūṭālīs*, for example, the earliest copy of which dates to the 8th/14th century, likewise became popular from Davānī onward, and was used most extensively by Mīr Dāmād in his *Qabasāt* (Pourjavady and Schmidtke, "An Eastern Renaissance?" pp. 263f.). Cf. the cognate quest of Giovanni Pico della Mirandola (d. 1494), Christian kabbalist extraordinaire, in his celebrated *Conclusiones*, as well as in his unpublished *Platonis Aristotelisque concordia*.

motion, had a soul and an intellect, and was immune from generation or corruption, whereas the sublunar region was formed of the four elements, moved in a linear fashion, and was subject to generation and corruption. Because of this, the processes in this celestial region were constant and could only be read using mathematical language, which was also rational because the celestial region had a correspondence with the intellects of human beings. This was the most important reason for astronomy having been included in the mathematical sciences in Mesopotamia, especially in the Sumerian-Babylonian-Chaldean heritage of science; in Mesopotamia this mathematical approach was in the form of arithmetic-algebra, whereas the Greeks demonstrated it more through the use of geometry. It is for the same reason that the Greeks regarded the sublunar region, which was a part of the natural world, as unreadable in terms of mathematics since it was a world of generation and corruption. As a matter of fact, the Aristotelian classification of theoretical philosophy into high, middle, and low corresponded to an ontological division: cosmologically, the sublunar region was the realm studied in the natural sciences, which were the lowest sciences (*al-ʿulūm al-adnā*), since they involved the lowest place; the celestial region was studied in the mathematical sciences, which were the middle sciences (*al-ʿulūm al-awsaṭ*), since they involved the middle region between the lunar sphere and the highest (*aṭlas*) sphere; and the realm beyond the highest sphere was studied in metaphysics or the theological sciences, which were the highest sciences (*al-ʿulūm al-aʿlā* or *ʿālīya*), since it was the realm of the Divine beyond the physical orbs ... [D]etermining the effects of the intelligent and active celestial region (*al-ağrām al-ʿulwiyya*) on the passive sublunar region (*al-ağsām al-suflīyya*) according to the ancient cosmology—in other words, astrology—required knowledge of this region's language. Forecasting future events, such as in the science of meteorology, is also related to the celestial region and taking necessary precautions necessitated the knowledge of astronomy, i.e. the language of the celestial region.⁴²

Yet Ibn Sīnā, who standardized the above schema in Arabic philosophy, breaks with it in an important way: he pointedly demotes astrology to the status of derivative natural science, together with alchemy, and dismisses them both as epistemologically invalid—an influential move that was to serve as model for

42 "The Samarqand Mathematical-Astronomical School," pp. 13 f.

encyclopedists and touchstone for anti-occultism polemics for centuries. In so doing, he was adopting, and subverting, the classificatory schema advanced by the great astrologer Abū Maʿšār Balḥī (d. 272/886) and his mentor, al-Kindī (d. 259/873), both of whom argued powerfully for the status of astrology as an aristotelian natural science. (Indeed, it was Abū Maʿšār's astrological writings in Latin translation that first introduced aristotelian natural philosophy to Europe.⁴³) As I argue, pro-occultist pushback can thus be partially measured through the gradual reclassification of certain occult sciences as mathematical, that is, trained on the supralunar realms of incorruptible, intelligible perfection as opposed to the elemental, sublunar realm of generation and corruption, where scientific precision is more elusive. In other words, the degree to which a given encyclopedist chooses to contest the Avicennan model by locating various occult sciences among the mathematical sciences may be considered a revealing index of neopythagorean-occultist sympathies. Not all occult sciences were susceptible of mathematicalization, of course, at least in the view of some: alchemy, for instance, would seem to be classifiable as a branch of physics, since it investigates the effects of bodies on bodies (*ḡasad fī ḡasad*).⁴⁴ Yet in the dominant Ḡābirian-Ḡildakian tradition even alchemy encompasses the supralunar realm by means of astrological and lettrist correspondences. Occult sciences like geomancy (*raml*), letter divination (*ḡafr*) or letter magic (*sīmiyā*) are more epistemologically slippery and polyvalent; they investigate the interaction of celestial-mathematical entities with sublunar elemental energies, or spirit-body relationships (*rūḥ fī ḡasad*),⁴⁵ and so constitute intersections of the natural and mathematical domains. The same consideration applies to astrology itself, which is predicated on the correspondence (*munāsaba*) of above to below along the Chain of Being. The choice that faced every encyclopedist was thus whether to privilege the celestial or the terrestrial aspects of a given occult science. I therefore argue further that the mathematicalization of certain occult sciences increased their appeal to both astronomer-cosmologists and metaphysicians, which in turn increased their appeal to royal patrons; and that this process, peculiar to the Persianate east,

43 See Lemay, *Abu Maʿshar*. The enthusiasm of the Kindī circle for astrology reflects the great popularity of that science in elite Abbasid society more generally as generated by the translation movement, as Gutas has shown (*Greek Thought*, pp. 1, 120, and *passim*). On the astrological theory of al-Kindī and Abū Maʿšār see Saif, *Arabic Influences*.

44 Thus Ibn Ḥaldūn's idiosyncratic definition (*Muqaddima*, pp. 489f.); see Melvin-Koushki, "Defending Geomancy."

45 Again, Ibn Ḥaldūn's definition. This is also the purpose of alchemy according to many practitioners (my thanks to Nicholas Harris for this observation).

must be considered an intrinsic part of the mathematical turn in astronomy on the one hand and the neopythagorean turn in metaphysics on the other.

The occult sciences that best reflect this development are three: astrology, geomancy and lettrism. (I here use *ʿilm al-ḥurūf wa-l-asmāʾ*, or lettrism, as an umbrella term incorporating a variety of magical and divinatory techniques based on the activation of the mathematical properties of the letters of the Arabic alphabet, including talismans and *ḡafr*, both of which are also associated with *sīmiyāʾ*.) By the early 9th/15th century all three were widely considered to be primarily mathematical sciences, and hence objects of intense scholarly interest and heavy royal patronage. Yet their shift in scientific status from natural to mathematical—not to mention the larger significance of this shift—has yet to be noted in any way by historians of Islamicate science, many of whom continue to be almost exclusively positivist or even triumphalist in orientation.⁴⁶ Despite their heavy representation in the sources and continuing popularity in much of the Muslim world, occult sciences like lettrism and geomancy are simply left unstudied as part of the history of science—intellectual dead ends, miasmic with superstition, that can be of interest only to anthropologists.⁴⁷ That the mathematicalization of astrology has never been flagged, however, is a grosser oversight, since the inextricability of astrology from astronomy in the Islamicate context has meant that astrology has come under the scrutiny of historians of science in a way no other occult science has; but the persistent failure to take the occult sciences seriously as an epistemologically interactive set has resulted in a total disjunction in the dominant narrative between astrology and various “postclassical” developments, including the mathematical turn, in astronomy, to say nothing of the other occult sciences. In essence, much of the relevant scholarship continues to proceed on the unstated but evident assumption that anti-occultist polemicists like Ibn Ḥaldūn won the debate by force of visionary brilliance and permanently relegated occultism to the shadows of unscience, at least in elite scholarly circles, thereby opening the floor to the “real,” almost “modern” scientists of the 9th/15th and 10th/16th centuries.⁴⁸ This assumption, of course, is violently ahistorical. Precisely the

46 Lemay, “L’Islam historique,” pp. 20–22; and see below.

47 Exceptions to this rule include most prominently the work of Emilie Savage-Smith, Charles Burnett and Sonja Brentjes. On the anthropological front, the best treatment to date of these occult sciences in their modern context is that by Alireza Doostdar (“Fantasies”).

48 Ibn Ḥaldūn’s objective in the *Muqaddima* is the reclassification of all occult sciences under the rubric of magic and divination, which have no scientific status but are exclusively products of innate psychic power. It is often forgotten in this context, however, that both

opposite happened: the reactionaries lost, the occultists won, and mathematical science flourished.

While Islamicate astrology continues to be studied almost exclusively in relation to its more “legitimate” parent, astronomy, then, and the other occult sciences continue to languish in latter-day obscurity, I suggest that one cannot properly evaluate the intellectual, cultural and political status of a science like astrology except in conjunction with its fellow mathematicalized occult sciences, particularly letter magic, letter divination and geomancy, which were increasingly synthesized with astrology to enrich their interpretive frameworks and extend their practical applications. And even alchemy was mathematicalized by the great Ġābirian adept ‘Izz al-Dīn Aydemir al-Ġildakī in the 8th/14th century (alchemy, in his universalizing reformulation thereof, encompassing and drawing on all the natural, mathematical and metaphysical sciences in its status as the supreme science),⁴⁹ then thoroughgoingly “astrologized” by his Ottoman heir ‘Alī Çelebi, the ‘new author’ (*müellif-i cedīd*) (in reality a group of anonymous authors), in the 11th/17th century.⁵⁰ Briefly put, that both geomancy and alchemy are often referred to as “terrestrial astrology” is no mere rhetorical flourish: this handle describes a fundamental epistemological continuity—one that must be taken more seriously than it has been to date.⁵¹

The balance of this article therefore investigates the process by which these occult sciences became interdependent mathematical sciences—and hence objects of intense patronage throughout the post-Mongol Persianate world—by surveying the shifting approaches to occultism on display in classifications of the sciences from the 4th/10th to the 10th/16th centuries, with emphasis on Persian encyclopedias, and with examples drawn from the 9th/15th-century Timurid-Aqquyunlu context and the 10th–11th/16th–17th-century Safavid-Mughal-Ottoman. It concludes by asking why these readily-documentable processes have eluded notice by historians of philosophy-science and suggesting directions for future research.

Ibn Ḥaldūn and Ibn al-Qayyim are even more concerned to delegitimize philosophy, being comparable in their fideistic position to William of Auvergne (d. 1249) (Melvin-Koushki, “Defending Geomancy”).

49 See below.

50 Artun, “Hearts,” p. 141. As Artun elsewhere notes, alchemy and lettrism also become increasingly inextricable, a notable feature of the Ġābirian tradition as a whole (*ibid.*, pp. 101f.).

51 On geomancy as terrestrial astrology see Melvin-Koushki, “Persianate Geomancy”; on alchemy as the same see Clulee, “*Astronomia inferior*.”

The Occult Sciences in Classifications of the Sciences

The aristotelian hierarchy outlined above was first proposed by Fārābī in his pathbreaking *Iḥṣāʾ al-ʿulūm* and fully worked out by Ibn Sīnā throughout his oeuvre. The first discusses only judicial astrology (*ʿilm al-ḥkām al-nuḡūm*) as an application of astronomy (*ʿilm al-nuḡūm al-taʿlīmī*), a mathematical science, both falling under the generic heading of *ʿilm al-nuḡūm*, the ‘science of stars’; and while he does not say so explicitly, Fārābī strongly implies that astrology belongs among the natural sciences, being a divinatory technique comparable to oneiromancy (*ʿibārat al-ruʾyā*), augury or ornithomancy (*zağr*) and fortunetelling (*ʿirāfa*).⁵² The Ṣayḥ al-Raʾīs then brings clarity to the astronomy-astrology relationship: as part of the quadrivium, astronomy is a mathematical science, but astrology is strictly a derivative natural science,⁵³ together with alchemy; and both occult sciences are to be rejected as epistemologically invalid.⁵⁴ So poorly does he regard arts occult, moreover, that Ibn Sīnā simply ignores them altogether in his *Dānišnāma-yi ʿAlāʾī*, the first Persian encyclopedia of the philosophical sciences.⁵⁵ The early Arabic administrative encyclopedia of Ibn Farīḡūn (d. after 344/955), *Ġawāmiʿ al-ʿulūm*, here slots comfortably into the developing Fārābian-Avicennan tradition; while it does treat of several occult sciences, including astrology, various forms of divination, magic, talismans and alchemy, it assigns them all to the category of natural philosophy.⁵⁶ (Ibn al-Nadīm (d. 380/990) is likewise more comfortable with the occult: his *Fihrist* devotes two chapters, of ten, to magical and divinatory techniques

52 *Iḥṣāʾ al-ʿulūm*, pp. 43f. Cf. al-Ḥwārazmī’s (d. c. 232/847) *Mafātīḥ al-ʿulūm*, whose chapter on astronomy, a mathematical science, includes a section on astrological terminology, but leaves the scientific status of astrology itself ambiguous.

53 Vesel, *Les encyclopédies*, p. 37.

54 See e.g. his *R. fi Aqsām al-ʿulūm al-ʿaqlīyya*, pp. 71–80. It should be noted, however, that the few natural sciences Ibn Sīnā lists in this brief work are all occult, with the exception of medicine, including astrology, physiognomy, oneiromancy, celestial magic, terrestrial magic and alchemy (p. 75).

55 Vesel, *Les encyclopédies*, p. 12. As Dimitri Gutas notes, the *Dānišnāma-yi ʿAlāʾī* is also the first philosophical work of any kind to be written in New Persian, and so Ibn Sīnā must be credited with singlehandedly fashioning modern Persian philosophical prose (*Avicenna*, p. 381). Given his anti-occultism, it is no coincidence that Ibn Sīnā is also hostile to neopythagorean doctrine. Gerhard Endress remarks in this connection: “[H]is is not a mathematician’s philosophy; and contrary to all of his predecessors, he leaves out all aspects of mathematical science where observational practice meets demonstrative method” (“Mathematics,” p. 142).

56 Vesel, *Les encyclopédies*, p. 24.

and alchemy respectively, and the second section of chapter seven treats of astrologers alongside astronomers and other specialists in the mathematical sciences.⁵⁷) Like Fārābī, Ibn Sīnā and Abū Maʿšar, Bīrūnī based his comprehensive *al-Taḥfīm fī awāʾil šināʿat al-tanǧīm* on the premise that judicial astrology, while predicated on a mastery of the mathematical sciences of geometry (*ḥandasa*), calculation (*ḥisāb*) and planetary theory (*ḥayʿat*), is itself an expressly natural (*ṭabīʿī*) science, and one driven by conjecture (*taḥmīn*) and the ability to perform divination (*kihāna*).⁵⁸

At the opposite extreme, the Iḥwān al-Šafāʾ proposed a very different and explicitly non-aristotelian hierarchy; as good neopythagoreans, they begin with the mathematical sciences as foundational to all else, proceed to logic, physics, metaphysics and the religious sciences, the hierarchy then culminating in occult sciences like alchemy, astrology and magic (along with medicine and *ʿilm al-taǧrīd*, the ‘science of emancipation’).⁵⁹ This approach was likely inspired in part by Ptolemy’s own reversal of the order of the sciences: the Alexandrian cosmographer-astrologer placed mathematics above both natural philosophy and metaphysics. As noted, however, despite the status of the *Rasāʾil* as lodestone for later occultism, it is the Avicennan epistemological hierarchy, featuring the mathematical sciences as mediators between heaven and earth, that became standard in much of subsequent high occultist theory.

After Ibn Sīnā’s *Dānišnāma-yi ʿAlāʾī*, the earliest Persian encyclopedias include three produced in the 6th/12th century. The *Nuzhatnāma* of Šāhmardān b. Abī l-Ḥayr Rāzī (fl. 513/1119) and the *Farruḥnāma* of Abū Bakr Muṭaḥhar Ġamālī Yazdī (fl. 580/1184) are administrative in orientation and scientifically amateurish, and while they treat of astrology and other occult sciences they do not employ the Avicennan model.⁶⁰ The *Yavāqīt al-ʿulūm va darārī al-nuǧūm*,

57 Lemay, “L’Islam historique,” pp. 24 f. In the tenth and final chapter on alchemy, Ibn al-Nadīm emphasizes Abū Bakr Rāzī’s (d. 313/925) status as a great defender of alchemy against the attacks of philosophers like al-Kindī (d. c. 259/873) and author of several alchemical works; he is also said to have asserted that no one can legitimately call himself a philosopher until he has mastered the Art (*al-Fihrist*, p. 541). On Rāzī as alchemist see Stroumsa, *Freethinkers*, pp. 88–92.

58 This, for instance, is the summary given by Nizāmī ʿArūzī (fl. 552/1157) in his *Čahār maqāla*, which features a full chapter (of four) on astrology and astrologers; see pp. 87–89. It should be remembered that Bīrūnī generally doubted the scientific validity of astrology, a position expressed in his *al-Qānūn al-Masʿūdī* (Brentjes and Morrison, “The Sciences,” p. 603).

59 *Epistles of the Brethren of Purity: On Magic I*, pp. 95 f.; de Callatāy, “Classification,” pp. 58–82; Saif, “From *Ġāyat al-ḥakīm*.”

60 Vesel, *Les encyclopédies*, pp. 27–31. Šāhmardān Rāzī, a secretary and accountant, also wrote

however, written by an anonymous Qazvini scholar for the local Seljuq governor Tuğril Tegīn Ġāzī Beg Alp Argū (d. 573/1177), is more sophisticated. Most notably, it proposes a binary classification scheme between the Islamic and pre-Islamic sciences, or 'latter-day sciences' (*'ulūm-i avāḥir*) and 'ancient sciences' (*'ulūm-i avā'il*), that became standard from this point forward; it is also the first encyclopedia to include sufism as an independent Islamic science, i.e., one on a par with hadith, jurisprudence or theology. At the same time, it classifies the occult sciences, here represented by oneiromancy (*ta'bīr-i ḥ'wāb*) and astrology (*nuḡūm*), under the rubric of natural philosophy (together with medicine and agriculture), followed by the mathematical sciences of geodesy (*masāḥat*) and calculation (*ḥisāb*), with a separate section at the end devoted to magical incantations (*ruqā u afsūn-hā*) and divinatory techniques (*fāl u zağr*).⁶¹

The first Persian encyclopedia *stricto sensu*, Faḥr al-Dīn Rāzī's (d. 606/1209) *Ġāmi' al-'ulūm* represents several other important firsts: it covers an unprecedented number of sciences (60, to be specific, hence its alternate title *Sitt-tīnī*), with emphasis on the philosophical; it reclassifies several occult sciences, including judicial astrology (*'ilm al-aḥkām*), geomancy (*'ilm al-raml*) and jinn magic (*'ilm al-'azāyim*),⁶² as mathematical sciences (together with the semi-occult science of magic squares), thereby initiating the trend this article investigates; and it is the first encyclopedia to include geomancy as a separate science. It follows the *Yavāqūt al-'ulūm* and other administrative encyclopedias in using an Islamic/pre-Islamic (or religious-literary/philosophical) binary classification system, with the philosophical sciences being loosely organized according to the Avicennan quadripartite scheme.⁶³ (Here "loose" is the operative word: the *Ġāmi' al-'ulūm* has little explicit internal structure, resembling a list more than an encyclopedia.⁶⁴) For the purposes of the present survey, it will be useful to list in full those sciences Rāzī includes under the twin rubrics of natural and mathematical:⁶⁵

Natural sciences (*'ilm al-tabī'īyyāt*): oneiromancy (*'ilm al-ta'bīr*), physiognomy (*'ilm al-firāsa*), medicine (*'ilm al-ṭibb*), surgery (*'ilm al-tašrīḥ*), phar-

an important Persian introduction to astrology, *Rawzat al-munaḡḡimūn*; see Storey, *Persian Literature*, vol. 2, pt. 1, p. 45, no. 81.

61 Vesel, *Les encyclopédies*, pp. 17–19; *Yavāqūt al-'ulūm*, pp. 67–78.

62 The 'science of incantations' (sg. *'azūmat*) often includes angel and planetary magic as well; Rāzī's definition is here unusually narrow.

63 Vesel, *Les encyclopédies*, pp. 35–39.

64 Biesterfeldt, "Encyclopedias," p. 97.

65 *Ġāmi' al-'ulūm* (*'Sittīnī*), pp. 253–447 (sciences 24–52).

macology (*‘ilm al-ṣaydana*), active properties (*‘ilm al-ḥavāṣṣ*), alchemy (*‘ilm al-iksīr*, *‘ilm al-kīmiyā*), gemmology (*‘ilm al-ḡavāhīr*), talismans (*‘ilm al-ṭilasmāt*), agriculture (*‘ilm al-filāḥa*), laundry (*‘ilm qal‘ al-āṣār*), veterinary medicine (*‘ilm al-bayṭara*), falconry (*‘ilm al-buzāt*).

Mathematical sciences (*‘ilm al-riyāẓiyyāt*): geometry (*‘ilm al-handasa*), geodesy (*‘ilm al-masāḥa*), mechanics (*‘ilm ḡarr al-aṣqāl*), war machines (*ālāt al-ḥurūb*), Indian arithmetic (*ḥisāb al-Hind*), mental calculation (*al-ḥisāb al-havā’i*), algebra (*al-ḡabr va-l-muqābala*), arithmetic (*‘ilm al-arīṣmāṭiqī*), magic squares (*‘ilm al-dād al-wafq*), optics (*‘ilm al-manāẓir*), music (*‘ilm al-mūsīqī*), astronomy (*‘ilm al-hay’a*), judicial astrology (*‘ilm al-aḥkām*), geomancy (*‘ilm al-raml*), jinn magic (*‘ilm al-‘azāyim*).

Thus, according to Rāzī, the natural occult sciences include oneiromancy, physiognomy, active properties, alchemy and talismans; the mathematical occult sciences include judicial astrology, geomancy and jinn magic (as well as magic squares, the basis of talismanic magic), which are closely linked with astronomy. That Rāzī conceives of the occult sciences as a distinct subcategory of natural and mathematical philosophy is suggested by the fact that he introduces geomancy to a new persophone audience with the opening injunction ‘Know that this science is one of the occult sciences’ (*bi-dān ki ‘īn ‘ilm az ‘ilm-hā-yi ḡarīb ast*), this while apparently considering sciences like astrology, alchemy and talismans to be too well-known as *ḡarīb* disciplines to require their explicit identification as such.⁶⁶ Despite its comprehensive picture of the intellectual scene in the Islamicate heartlands at the turn of the 7th/13th century and its cutting-edge status, however, it should be noted that the *Ġāmi‘ al-‘ulūm*’s treatment of many sciences is rather abbreviated; its chapter on geomancy, for instance, is too rudimentary to be useful to a would-be practitioner, suggesting that this occult science had not yet fully entered the scholarly mainstream in the Persianate world.⁶⁷ Nevertheless, Rāzī represents a key player in the burgeoning pro-occultism movement as author of treatises on physiognomy, astrology, geomancy, chiromancy and oneiromancy. In particular, his *al-Sirr al-maktūm*, on talismanic and jinn magic, enjoyed immense popularity as a manual of occultism in subsequent centuries, including in Mughal India.⁶⁸ Of

66 *Ġāmi‘ al-‘ulūm*, p. 432.

67 *Ġāmi‘ al-‘ulūm*, pp. 432–437. On the development of the eastern geomantic tradition see Melvin-Koushki, “Persianate Geomancy.”

68 It is the only eastern occultist text that the poorly informed Ibn Ḥaldūn is aware of; he also

similar significance is the fact that he also advanced an incisive critique of aristotelian form-matter analysis, proposing an alternative approach to the study of natural phenomena.⁶⁹ It is therefore unsurprising that he would be responsible for the first push to mathematicalize certain occult sciences, this both by virtue of his occult-philosophical outlook and as an expression of his rivalry with Ibn Sīnā.⁷⁰

Later in the 7th/13th century, Faḥr al-Dīn Rāzī's move to reclassify astrology as a mathematical science was repeated in the Persian administrative encyclopedia of Šams al-Dīn Dunaysirī, *Navādir al-tabādur li-tuḥfat al-bahādur*, completed in 682/1283 for an unnamed amir at the Qarāḥiṣār fortress.⁷¹ Like the *Nuzhatnāma* and *Farruḥnāma*, the *Navādir al-tabādur* is scientifically amateurish, but unlike them gives pride of place to astronomy and astrology: it devotes two full chapters, out of 12, to these sciences, and clearly classifies them both as mathematical.⁷² Moreover, in the last chapter, devoted to taxonomizing the miraculous, Dunaysirī significantly treats magic (*siḥr*, *nīranġāt*) as being on a par with prophetic and saintly miracles.

Rāzī's revisions did not go unchallenged for long, however; Quṭb al-Dīn Šīrāzī, the important illuminationist philosopher-astronomer and member of the Maragha school, powerfully reasserted Ibn Sīnā's schema in his Persian *Durrat al-tāġ li-ġurraṭ al-dabbāġ*, written sometime between 693–705/1294–

believes the 4th/10th-century *Ġāyat al-ḥakīm* of Ps.-Maġrītī (rather Maslama al-Qurṭubī) to be the final word on magic, and that no one has written on it since (*Muqaddima*, p. 483). See Vesel, "The Persian Translation," pp. 14–22.

69 Ibrahim, "Freeing Philosophy."

70 Gutas, "Certainty, Doubt, Error," p. 284: "In the case of Fakhraddīn ar-Rāzī, the objective seems to be even more base than that [of Maimonides' refutation of Galen]; he was extremely jealous of Avicenna's prominence and simply wished to promote his own views in order to gain the fame Avicenna enjoyed. Fakhraddīn is not the only scholar to have reacted in this way to Avicenna—it seems that Avicenna did have this effect on people."

71 Ed. Muḥammad-Taqī Dānišpažūh and Īraġ Afšār; Vesel, *Les encyclopédies*, pp. 31–33.

72 Specifically, chapter two ('On Mathematical Science' (*andar 'ilm-i riḡāzī*)) of the *Navādir al-tabādur* is subdivided into six sections: 1) on the states of the celestial sphere; 2) on recognizing the planets (*kavākib-i mutaḥayyira*); 3) on the nature and constitution of the seven planets; 4) on the signification of the twelve houses; 5) on the stellar aspects; 6) on the states of the 28 lunar mansions (with charts of correspondences). Chapter three ('On Various Kinds of Useful Astronomical-Astrological and Geometrical Information' (*andar čand naw' az favāyid-i nuġūmī u handasī*)) likewise has six sections: 1) on the science of the astrolabe; 2) on planetary invocations; 3) on planetary ephemerides; 4) on determining the Ascendant of the year's revolution (*ṭālī-i taḥvīl-i sāl*); 5) on determining the Ascendant of natiivities (*ṭālī-i mavālid*); 6) on the science of the *tasyrāt*.

1306 for Amir Dabbāğ, ruler of Gilan.⁷³ Hailed as a pivotal work in the Islamicate encyclopedic tradition (Paul Heck observes, for example, that therein Šīrāzī “might have come closer than any other to succeeding in creating a unified intellect-based classification of knowledge”⁷⁴), the *Durrat al-tāğ* reclassifies all occult sciences, including judicial astrology, physiognomy, oneiromancy, alchemy, talismans and *nīranğāt*, as branches of natural philosophy, together with medicine and agriculture. (Helpfully for our purposes here, Šīrāzī defines talismans (*‘ilm-i ṭīlismāt*) as a form of celestial magic that conjuncts celestial virtues with the virtues of certain terrestrial bodies (*tamzīğ-i quvā-yi samāwī bi quvā-yi ba’zī aḡrām-i arzī*) to produce a power (*quvvatī*) that gives rise to an unusual action in this world (*mabda’-i fi’lī ġarīb dar īn ‘ālam*), and defines *nīranğāt* as a form of terrestrial magic that conjuncts terrestrial virtues with one another to produce a similar power productive of an unusual action.⁷⁵) At the same time, his treatment of such sciences is much fuller than Ibn Sīnā’s, and he does not declare any of them epistemologically invalid in the manner of his august predecessor, whom he regarded rather dismissively.⁷⁶ While he resists Rāzī in particular, then, Šīrāzī’s illuminationist perspective is likely responsible for his occultism-friendly position as expressed in the *Durrat al-tāğ*—indeed, Suhrawardī (d. 587/1191) himself is renowned as a practicing occultist in the later Islamicate tradition.⁷⁷ More importantly, Šīrāzī’s critical undercutting of the Avicennan epistemological hierarchy laid the groundwork for the later mathematical turn in astronomy: he elsewhere declares that only mathematical proofs can produce certain knowledge, unlike those of natural philosophy or metaphysics, and hence mathematical astronomy (*‘ilm al-hay’a*) is the supreme science (*ašraf al-‘ulūm*).⁷⁸

The *Nafāyis al-funūn fi ‘arāyis al-‘uyūn* of Šams al-Dīn Muḥammad Āmulī (d. 753/1352), sometime *mudarris* at Sultaniyya under Öljeytū (r. 703–716/1304–

73 Ed. Muḥammad Miškāt.

74 “The Hierarchy,” p. 53 n. 93; see Vesel, *Les encyclopédies*, pp. 13–15; Bakar, *Classification*, pp. 227–262.

75 *Durrat al-tāğ*, pp. 155 f. Here Šīrāzī is restating Ibn Sīnā’s definition of the same; see e.g. the latter’s *R. fi Aqsām al-‘ulūm al-‘aqlīyya*, p. 75.

76 Pourjavady and Schmidtke, “An Eastern Renaissance?” p. 252.

77 See e.g. Walbridge, “Works”; idem, *Wisdom*, pp. 46 f.

78 *Nihāyat al-idrāk fi dirāyat al-aflāk*, Süleymaniye MS Ayasofya 2733/2, ff. 1^b–2^a; Fazlıoğlu, “Between Reality and Mentality,” pp. 7 f. Šīrāzī does not dispense with metaphysical or natural philosophical principles, it should be noted, but rather creates “a hierarchy for the value of proof in different disciplines that privileges mathematics” (ibid., p. 8).

1316), represents another watershed in the Islamicate encyclopedic tradition.⁷⁹ It is the most extensive and polished *taṣnīf al-ʿulūm* work produced in either Arabic or Persian up to the mid-8th/14th century, and as such came to serve as the primary model for subsequent Persian encyclopedias.⁸⁰ The *Nafāyis al-funūn* has as its own model Faḥr al-Dīn Rāzī's *Ġāmiʿ al-ʿulūm*, but it is far more complete and organized than that pioneering work. Most significantly, Āmulī, unlike Rāzī, questions the Avicennan hierarchy of sciences so forcefully reiterated by Quṭb al-Dīn Šīrāzī, at times even challenging the latter directly by counterposing what he asserts to be a more consistent classification system and moving toward a new conception of knowledge.⁸¹ At the same time, he concurs with Šīrāzī in classifying all occult sciences, including alchemy, letter magic (*sīmīyā*), oneiromancy, physiognomy and astrology, as derivative natural sciences—with one important exception: geomancy. We saw above that Rāzī, the first encyclopedist to treat of this form of “terrestrial astrology,” explicitly classified it as an applied mathematical science associated with astronomy; Āmulī agrees with him on this point, while also dissociating the science from astrology on the one hand and all other divinatory sciences on the other. Though his motive for making it is not clear, Āmulī's move here may be considered an index of geomancy's increase in prestige in Iran during the Ilkhanid period. (I have shown elsewhere that the Persian geomantic tradition properly begins with Naṣīr al-Dīn Ṭūsī (d. 672/1274) at the instance of his Mongol patrons and burgeons impressively from there, reaching a crescendo of popularity among scholarly and ruling elites in the runup to the Islamic millennium (1000/1592) and persisting in the Persianate world to the present.⁸²) Moreover, at 20 pages Āmulī's discussion of geomancy is considerably fuller than his model's and detailed enough to serve as a basis for elementary practice.

As suggested above, the *Nafāyis al-funūn* builds the various features developed by his predecessors into a cohesive and elegant structure. Like the *Ġāmiʿ al-ʿulūm*, it employs the now-standard basic division between the Islamic or latter-day sciences (*ʿulūm-i avāḥir*) and the philosophical or ancient sciences (*ʿulūm-i avāʾil*), and covers an unprecedented number of both. Like the *Durrat al-tāğ*, it employs a highly structured Avicennan schema that first treats of the roots of philosophy (*uṣūl-i ḥikmat*) (logic, first philosophy, metaphysics, physics) and the roots of mathematics (*uṣūl-i riḳāzī*) (the quadrivium: geometry (*uṣṭuquṣṣāt*, *handasa*), astronomy (*aṣṭarunūmā*, *nuğūm*), arithmetic

79 Ed. Abū l-Ḥasan Šaʿrānī and Ibrāhīm Miyānġi.

80 Vesel, *Les encyclopédies*, pp. 39–42.

81 Vesel, *Les encyclopédies*, p. 41.

82 See Melvin-Koushki, “Persianate Geomancy.”

(*arīṣmāṭiqī*, *a'dād*), music (*mūsīqī*)), then proceeds to discuss various branches first of the natural sciences (ten) and then of the mathematical sciences (16) as follows:

Branches of natural science (*furū'-i ṭabī'ī*): medicine (*'ilm-i ṭibb*), alchemy (*'ilm-i kīmīyā*), letter magic (*sīmīyā*), oneiromancy (*'ilm-i ta'bīr*), physiognomy (*'ilm-i firāsāt*), judicial astrology (*'ilm-i aḥkām-i nuḡūm*), active properties (*'ilm-i ḥavāṣṣ*), physiological professions (*'ilm-i ḥiraf al-ṭabī'a*) (these including the two divinatory techniques of scapulomancy (*aktāf*) and spasmatomancy (*iḥtilāğ-i a'zā'*)),⁸³ breath control (*'ilm-i dam*), meditation and visualization (*'ilm-i vahm*) (*dam* and *vahm* being Indian yogic sciences).

Branches of mathematical science (*furū'-i ri'yāzī*): planetary theory (*'ilm-i ḥay'at*);⁸⁴ optics (*'ilm-i manāẓir*); the Middle Books (*mutavassīṭāt*); calculation (*'ilm-i ḥisāb*); algebra (*'ilm-i ḡabr u muqābala*); geodesy (*'ilm-i masāḥa*); constellations (*'ilm-i šuvar-i kavākib*); star tables, ephemerides and astrolabes and other observational instruments (*'ilm-i arqām u a'māl-i zīğ u taqvīm u aṣṭurlāb u ālāt-i raṣadī*);⁸⁵ administrative geography (*'ilm-i masālik u mamālik*); magic squares (*'ilm-i vaḥq-i a'dād*); geomancy (*'ilm-i raml*); mechanics and wondrous devices (*'ilm-i ḥiyāl*);⁸⁶ games of strategy (*'ilm-i malā'ib*).

What makes the *Nafāyis al-funūn* truly pivotal, however, is its status as the first encyclopedia to register a) the rise of sufism to sociopolitical hegemony, and b) the sanctification of occultism. As described above, the second process stemmed from the fusion of sufism and occultism primarily under the

83 According to the *Nafāyis*, the physiological professions are six: veterinary medicine (*bayṭara*), falconry (*bāzdārī u murğšikārī*), laundry (*raf'-i āsār-i ḡāma az ālūdagī*), agriculture (*filāḥat*), scapulomancy (*aktāf*), spasmatomancy (*iḥtilāğ-i a'zā'*).

84 It should be noted that Āmulī uses the term *nuḡūm* for astronomy as part of the quadrivium and *ḥay'at* for one of its branches.

85 Āmulī valorizes this category in particular, declaring these to be the most useful of all the mathematical sciences (*Nafāyis al-funūn*, vol. 3, p. 459).

86 Āmulī includes under the rubric of *ḥiyāl*, defined here as the production of wondrous (*ḡarīb*) movements and the construction of automata, hydraulics (*naql-i miyāh*), mechanics (*ḡarr-i aṣqāl*) and war machines (*ālāt-i ḥurūb*) (*Nafāyis al-funūn*, vol. 3, p. 557). Cf. John Dee's definition of the same, under *thaumaturgike*, as "the straunge Selfmoyving" (*Mathematicall Praeface*, sig. A.j).

banner of Ibn ‘Arabī and al-Būnī during the course of the 7th–8th/13th–14th centuries, with a focus on the newly “suficized” science of letters (*‘ilm al-ḥurūf*) as the science of the saints (*‘ilm al-awliyā’*) *par excellence*: equal parts magic, divination, mystical exegesis and cosmological speculation, and conferring control over the physical, imaginal and metaphysical realms. Āmulī flags these twin developments by first elevating the science of sufism (*‘ilm-i taṣawwuf*) to the status of supreme Islamic science, equal in importance to all the other religious sciences (including jurisprudence, hadith and theology) combined, then designating lettrism the supreme sufi science.⁸⁷ At the same time, he retains the category of *sīmīyā*, letter and talismanic magic, as an applied natural science, further classifying it as one of the ‘Semitic sciences’ (*‘ulūm-i sāmīyya*)—i.e., positing a connection to Hebrew kabbalah.⁸⁸ Yet even there he stipulates that proficiency in *sīmīyā* is predicated on, among other things, a mastery of astronomy (a mathematical science) and astrology (a natural science):

The 15th [and final condition for the successful performance of a letter-magical operation] is thorough proficiency in *‘ilm-i nuḡūm*, including the natures of the planets, signs of the zodiac and mansions of the moon, the natures of the celestial degrees and the auspicious and inauspicious aspects, such that one is able to undertake an operation knowing which planet is governing it and the things associated therewith, like days [of the week], configurations (*hay’āt*), colors, shapes, foods, beverages, items of clothing, incenses, etc.; one can thus employ all of them [in the operation] and thereby strengthen its effect.⁸⁹

Āmulī’s sophisticated and nuanced classification here thus signals the emergence of lettrism as a simultaneously Islamic, natural and mathematical science—that is to say, a *universal* science—and a defining feature of the religio-intellectual landscape of the Islamicate heartlands from the mid-8th/14th century onward. Strangely enough, the best testament to the incontrovertible nature of this development is provided by Ibn Ḥaldūn himself: despite the Tunisian historian’s obsessive and reactionary anti-occultism, he joins Āmulī

87 *Nafāyis al-funūn*, vol. 2, pp. 91–110.

88 *Nafāyis al-funūn*, vol. 3, p. 183. Ibn al-Akfānī (on whom see below) gives an etymology of the term *sīmīyā* as deriving from the Hebrew *šem Yah*, ‘the name of God,’ indicating the science’s association with the divine names as loci of magical power (*Iršād al-qāṣid ilā asnā l-maqāṣid*, p. 51).

89 *Nafāyis al-funūn*, vol. 3, p. 191.

in synthesizing in his *Muqaddima* the categories of sufism and *sīmiyā*.⁹⁰ Unlike Āmulī, however, he makes *‘ilm al-ḥurūf* a subset of *sīmiyā*, while also emphasizing both its basis in neoplatonic philosophy and its association with Ibn ‘Arabī and al-Būnī:

The science originated in Islam after some time of (its existence) had passed. When the extremist Sufis appeared, they turned to the removal of the veil of sense perception, produced wonders, and were active in the world of the elements. At that time, they wrote systematic works on (Sufism) and (Sufi) terminology. They believed in the gradual descent of existence from the One. They believed that verbal perfection consists in helping the spirits of the spheres and the stars (through words). The natures and secrets of the letters are alive in the words, while the words, in turn, are correspondingly active in the created things. The created things have been moving in the different stages of (creation) and telling its secrets since the first creation. These (Sufi beliefs) caused the science of the secrets of the letters to originate. It is a subdivision of the science of *sīmiyā* ‘letter magic.’ It is an unfathomable subject with innumerable problems. Al-Būnī, Ibn al-‘Arabī, and others in their wake wrote numerous works on it. These authors assume that the result and fruit of letter magic is that the divine souls are active in the world of nature by means of the beautiful names of God and the divine expressions that originate from the letters comprising the secrets that are alive in the created things.⁹¹

Enter Sayyid Ḥusayn Aḥlāṭī: like Ibn Ḥaldūn an immigrant to Cairo, the persophone Syrian Kurdish *magus* epitomized this new brand of Ibn ‘Arabian-Būnian lettrism so completely that he was hailed by his disciples and their intellectual heirs as singlehandedly launching the great renaissance of occultism that swept the Persianate world in the early 9th/15th century, with important consequences for both metaphysical and mathematical-scientific thought. While his own surviving writings are devoted exclusively to practical matters relating to letter magic and letter divination (that is to say, are essentially Būnian), those of his two most important and prolific students—‘Abd

90 Ibn Ḥaldūn, infamously, is very poorly informed of matters beyond North Africa, and especially political and intellectual developments in the Persianate east, so there is little question of his having firsthand familiarity Āmulī’s work.

91 *The Muqaddimah*, vol. 3, pp. 171 f.

al-Raḥmān al-Bisṭāmī (d. 858/1454) in Anatolia and Ibn Turka in western Iran—reformulated lettrism into an explicitly universal science along the lines suggested by Āmulī several decades prior.⁹² But they go beyond him: both al-Bisṭāmī and Ibn Turka decouple lettrism from sufism with the object of further valorizing the first. While al-Bisṭāmī posits lettrism and sufism as separate but equally supreme Islamic sciences that lead to the same goal,⁹³ however, Ibn Turka is more radical. The Isfahani judge asserts lettrism to be the sole universal science in its status as the only means for actualizing *walāya*, on the one hand, and the only valid form of metaphysics on the other; it thereby trumps both sufism and peripatetic-illuminationist philosophy, which are valid within their own remit but no more than preparatory for the final stage of intellectual-spiritual ascent. Lettrism thus represents for Ibn Turka the definitive confluence of the Islamic and pre-Islamic legacies: it is simultaneously the science of the Imams and the ancients. As an Imamic legacy associated with ‘Alī b. Abī Ṭālib and Ġa‘far al-Šādiq in particular, lettrism confers upon the adept control of history and the cosmos; as a Solomonic legacy associated with Pythagoras, Plato and Ps.-Aristotle (i.e., Plotinus) in particular, lettrism represents the culmination of neoplatonic-neopythagorean philosophy. That is to say, lettrism represents a *coincidentia oppositorum*, the explicit basis of all occultist theory during this period: it alone resolves the binary posited by Muslim encyclopedists for centuries between the *‘ulūm-i avāḥir* and the *‘ulūm-i avā’il*, creating a single, unbroken intellectual-spiritual genealogy stretching back to Adam himself—the *prisca theologia*, the *philosophia perennis*.⁹⁴

Although Ibn Turka develops these themes in a number of works devoted to theoretical and applied lettrism, some written for members of the Timurid ruling elite, they are most fully worked out in his *Book of Inquiries* (*K. al-Mafāḥiṣ*), a definitive systematization of Aḥlāṭī’s teachings along Ibn ‘Arabian and Ḡābirian lines best described as the first *summa* of Islamic neopythagoreanism. This

92 On ‘Abd al-Raḥmān al-Bisṭāmī see Fleischer, “Ancient Wisdom”; Fazlhoğlu, “İlk Dönem Osmanlı İlim”; Gril, “Ésotérisme.” On Akhlāṭī see Binbaş, *Intellectual Networks*, pp. 114–140; Melvin-Koushki, “The Quest,” pp. 47–57 and *passim*.

93 Al-Bisṭāmī does emphasize, however, that lettrism constitutes the more scientific or methodical route: “In his sociology, the noble Ḥurūfīs (*sādat al-ḥarfīyya*) and noble Mystics (*sādat al-ṣūfīyya*) are parallel, the latter attaining divine knowledge through inspiration, the former through scientific praxis” (Fleischer, “Seer to the Sultan,” p. 293).

94 Melvin-Koushki, “The Quest,” pp. 321–324. On these terms and their centrality in early modern Christianate occultism, as well as their divergent connotations, see Hanegraaff, *Esotericism*, pp. 5–12.

work treats of the meanings of the letters according to their three forms, numerological (*iḥsā'ī*), symbolical (*kitābī*) and phonological (*kalāmī*), as well as the letters as they are in themselves (*fī anfusi-hā*).⁹⁵ As the author elsewhere states, knowledge of these three forms is the sole preserve of the companions and true heirs of the Prophet (*aṣḥāb al-ḥātam wa-warathatu-hu*)—i.e., those *men of might and vision* (Q 38:45) that he assigns to the seventh and highest rank in his intellectual hierarchy and identifies in the first place with the Imams and the saintly sages of Greece, disciples of the prophets of God (especially Solomon). By this definition, then, the *Maḥāfiḥ* must be considered a manual for attaining to a simultaneously 'Alī-like and Pythagoras-like status.⁹⁶

Ibn Turka states that his primary purpose in this work is to demonstrate the roots of all manifestation in the One and schematize the mechanics of multiplicity's derivation therefrom. This information, in turn, will allow the adept to manipulate the letters—the uncreated, all-creative matrices through which the One self-manifests—to access and control every epistemological and ontological level of the cosmos, thus constituting a continuum from rarefied letter theory to purely practical letter magic and letter divination. The supreme dignity of its object necessarily renders lettrism the supreme science:

The subject of the science we have here in view is the One (*al-wāḥid*) insofar as it is one, regardless of the form in which it manifests in all the variety of its significations. The all-pervasive, all-encompassing nature of One with respect to existence being obvious, this science is therefore necessarily superior to all other sciences by an order of magnitude.⁹⁷

It enjoys this status, moreover, precisely because it is the most *mathematical* of all the sciences:

95 It is here highly significant that in his challenge to philosophy Ibn Turka seeks to displace *wuḡūd* (or the illuminationists' *nūr*) with *ḥurūf* in all respects; thus the concepts that would be so central to Mullā Ṣadrā's system, the ontological primacy of being (*aṣālat al-wuḡūd*) and the graded or transcendent modulation of being (*taṣkik al-wuḡūd*) (see Rizvi, *Mullā Ṣadrā*), are with Ibn Turka *aṣālat al-ḥurūf*, the ontological primacy of the letters, and *taṣkik al-ḥurūf*, the modulation of the letters in written, verbal, mental and extra-mental form. On this theme see my forthcoming "Of Islamic Grammatology: Ibn Turka's Lettrist Metaphysics of Light."

96 Melvin-Koushki, "The Quest," pp. 315–320, 330 f.

97 *K. al-Maḥāfiḥ*, MS Majlis 10196, f. 53^b; see Melvin-Koushki, "The Quest," p. 331.

Behind this gate [i.e., the new lettrist theory here proposed] are the treasuries of the Seal [Muḥammad's] glory and the protected space of his intimacy which contain necklaces of precious jewels (*'uqūd farā'id al-ġawāhir*) and all else laid there in store. [Muḥammad] collected all this and provisioned therewith his son [ʿAlī b. Abī Ṭālib], the Seal of sacral power (*walāya*) and standard-bearer of understanding and guidance. These necklaces (*'uqūd*) are numerical knottings (*al-'uqūd al-'adadiyya*), the spiritual-intellectual form of the Book that was sent down from the highest Pen to the noble Tablet. Number (*'adad*), then, is the best means of acquiring sciences of great benefit and numerous as grains of sand, the primordial mine preserving the gems [at the core] of all the standard and mainstream sciences ... Let the researcher therefore set aside his various misconceptions and inquire into the matter of number, for it is the fountainhead of all the sciences, the quarry of all realities, an ocean of insights both manifest and occult.⁹⁸

In short, Ibn Turka here completes the sanctification of occultist neopythagoreanism, a process begun by the Iḥwān al-Ṣafā'—their *Rasā'il* likewise founded on the premise that “the science of number is the root of the sciences, the essence of wisdom, the foundation of knowledge and the [principal] element of all things”⁹⁹—, by associating it definitively with ʿAlī.¹⁰⁰ According to Ibn Turka's

98 *K. al-Mafāḥiṣ*, MS Majlis 10196, ff. 52^a, 56^b; Melvin-Koushki, “The Quest,” pp. 338, 344. Note that *'adad*, translated here as ‘number,’ is also the standard term for arithmetic as part of the quadrivium.

99 *Rasā'il*, vol. 1, pp. 21 f.; trans. Endress, “Mathematics,” p. 133.

100 It is here significant that Ibn Turka breaks with Ibn ʿArabī's identification of the *ḥātām al-walāya al-muṭlaqa/al-ʿamma* as Jesus, in this appearing to follow the Shi'i mystical philosophers ʿAlī b. Sulaymān al-Baḥrānī (d. c. 669/1271), Maytham b. Maytham al-Baḥrānī (d. 699/1300) and Ḥaydar Āmulī (d. after 787/1385), who similarly awarded this status to ʿAlī as part of their project to synthesize Ibn ʿArabī's theory with Twelver theology (al-Oraibi, “Rationalism,” pp. 333 f.). While the imamophilic aspect of Ibn Turka's occult-philosophical project led scholars like Henry Corbin and his heirs to reflexively assume that Ibn Turka identified as Shi'i, however, his writings express a very different motive: to construct a new lettrist metaphysics epistemologically superior to both sufism and philosophy and explicitly perennialist in cast, to this end positing the Imams and the ancients as equally potent sources of occult, and especially lettrist, knowledge. Cf. Ḥāfiẓ Raġab al-Bursī (d. 813/1411), Ibn Turka's Twelver lettrist contemporary, whose *K. Maṣāriq anwār al-yaqīn fī ḥaqā'iq asrār amūr al-mu'minīn* represents an extremist sufi-Shi'i approach to the science; it is concerned to demonstrate the occult supremacy of the Fourteen Infallibles and develop the doctrine of the *walāya* of the 12 Imams on lettrist bases, going so far

reclassification thereof, moreover, lettrism, as a neopythagorean-neoplatonic mode of knowledge intrinsic to the prophetic-philosophical legacy throughout history and operative on all levels of reality, is alone capable of serving as a universal science organizing all of human knowledge under its aegis. To borrow Vittoria Perrone Compagni's description of the not dissimilar objective of Heinrich Cornelius Agrippa (d. 1535) in his *De Occulta Philosophia Libri Tres*, Ibn Turka "proposes a total reorganization of [lettrism] as an umbrella science which, by gathering under a single roof all the cognitive data collected in the various fields of scientific research, would guarantee the effectiveness of each branch of research and make explicit its potential for acting upon reality."¹⁰¹

Ibn Turka's lettrism, running the gamut from metaphysics to mathematics to magic (a natural science), is thus immediately comparable to the Christian kabbalah first developed several decades later by European Renaissance thinkers like Giovanni Pico della Mirandola (d. 1494), who posited it as the only discipline capable of uniting all conceptual opposites, of reconciling religion and philosophy, of marrying Plato and Aristotle. Most strikingly, it also stands as rather precise cognate to John Dee's (d. 1608) celebrated "cabala of the real," the primary methodology of that *archemastrie*—i.e., supreme science—demonstrated in the Welsh occultist's *Monas Hieroglyphica* (1564) and lionized in his *Mathematicall Praeface* (1570). In the latter work Dee defines his new universal, kabbalistic science as follows:

This Arte, teacheth to bryng to actuall experience sensible, all worthy conclusions by all the Artes Mathematicall purposed, & by true Naturall Philosophie concluded: & both addeth to them a farder scope, in the termes of the sames Artes, & also by his propre Method, and in peculier terms, procedeth, with helpe of the foresayd Artes, to the performance of complet Experiēces, which of no particular Art, are hable (Formally) to be challenged ... *Science* I may call it, rather, then an Arte: for the excellency and Mastershpy it hath, over so many, and so mighty Artes and Sciences. And bycause it procedeth by *Experiences*, and searcheth forth the causes of Conclusions, by *Experiences*: and also putteth the Conclusions them selves, in *Experience*, it is named of some, *Scientia Experimentalis*. The *Experimentall Science* ... [T]he Astronomer, and the

as to declare them eternal principles and 'Ali divine (Melvin-Koushki, "The Quest," p. 218 n. 184).

101 See the introduction to her edition of *De Occulta Philosophia Libri Tres*, p. 16.

Opticall Mechanicien, put some thynges in *Experience*: but neither, all, that they may: nor yet sufficiently, and to the utmost, those, which they do. There, then, the *Archemaster* steppeth in, and leadeth forth on, the *Experiences*, by order of his doctrine *Experimentall*, to the chief and finall power of Naturall and Mathematicall Artes.¹⁰²

(It bears noting here that lettrism too, like astrology before it, was redefined as a *tağribī* or experiential science during the 9th–10th/15th–16th centuries.¹⁰³) Interestingly enough, Dee further identifies the specifically Arabic tradition of natural magic as being of particular importance for the attainment of *archemastrie*: “To this Science, doth the *Science Alnirangiat*, great Service.”¹⁰⁴ It is further suggestive in this connection that Dee, like Faḥr al-Dīn Rāzī (and Rāzī’s near-contemporary Roger Bacon (d. 1292)¹⁰⁵), explicitly recategorizes astrology as a mathematical science, and for similar reasons:

Of Astrologie, here I make an Arte, severall from *Astronomie*: not by new devise, but by good reason and authority: for, Astrologie, is an Arte Mathematicall, which reasonably demonstrateth the operations and effectes, of the naturall beames, of light, and secrete influence: of the Sterres and Planets: in every element and elementall body: at all times, in any Horizon assigned. This Arte is furnished with many other great Artes and experiences: As with perfecte *Perspective*, *Astronomie*, *Cosmographie*, *Naturall Philosophie* of the 4. Elementes, the Arte of Graduation, and some good understāding in *Musike*: and yet moreover, with an other great Arte [i.e., *Archemastrie*]¹⁰⁶

102 *Mathematicall Praeface*, sig. A.iiij.

103 As ‘Alī Ṣafī Kāshifī (d. 939/1533), son of the great Timurid polymath and occultist Ḥusayn Vā’iz Kāshifī (d. 910/1505), asserts in his lettrist treatise *Ḥirz al-amān min fitan al-zamān*, which cites both Aḥlāṭī and Ibn Turka: ‘Experiential assaying (*tağriba*) aimed at manifesting the effects and properties of the letters has often been successfully repeated; and experience is among the forms of proof that produce certainty (*tağriba yakī az isbāt-i yaqīn ast*)’ (see Melvin-Koushki, “The Quest,” p. 279). Much earlier, Abū Ma’sar had influentially asserted astrology to be a science founded on analogical reasoning (*qiyās*) and experience (*tağriba*), the latter entailing observation (*mu’āyana*) and direct discovery (*al-wuḡūd al-ḥādir*) (Burnett, “Certitude,” pp. 210, 212).

104 *Mathematicall Praeface*, sig. A.iiij. As noted above, Quṭb al-Dīn Šīrāzī defines *nīranḡāt* as terrestrial magic.

105 See Easton, *Roger Bacon*, p. 7; Hackett, “Reception,” pp. 149–162.

106 *Mathematicall Praeface*, sig. b.iii.

Simon Forman (d. 1611), Dee's contemporary, did his fellow anglophone occultist one better by incorporating even alchemy as a mathematical science, then fusing it with both astrology and kabbalah.¹⁰⁷

There can be no question of Arabic (or Persian)-to-Latin influence here, of course (quite apart from the fact that Quran-centric lettrism, unlike the "secular," natural-philosophical magic of the *Ġāyat al-ḥakīm*, was unadaptable to a Christian context); I invoke these examples here merely to flag the double standard that still prevails among historians of science, whereby Pico's or Dee's obsession with kabbalah—or Kepler's (d. 1630) self-identification as a neopythagorean¹⁰⁸—heralds the modern mathematization of the cosmos but Ibn Turka's obsession with lettrism, kabbalah's twin, heralds but Islamic decadence: for Islam produced no Newton. Most perniciously, this double standard elides a major problematic in global history of science and philosophy. Triumphalist teleologies aside, that is, it is remarkable that, in the absence of direct contact, the quest for a universal science was universally pursued along neopythagorean-occultist lines throughout the Islamo-Christianate world during the "early modern" period—a trend that became mainstream significantly earlier in the Islamicate context.

Already in the early 8th/14th century, by way of example, Aydemir al-Ġildakī had beaten out both Ibn Turka and Christian kabbalists like Forman in his own reformulation of alchemy, rather than lettrism or astrology, as the universal science incorporating all other natural, mathematical and metaphysical sciences, including in the first place astronomy (*ʿilm al-falak*), arithmetic, geometry, medicine and lettrism.¹⁰⁹ In the preface to his *al-Miṣbāḥ fī asrār ʿilm al-miftāḥ*, for instance, al-Ġildakī asserts the following:

I hold that among the principal mathematical sciences (*uṣūl ʿulūm al-taʿālīm*) that are a prerequisite for the Divine Art (*al-ṣināʿa al-ilāhiyya*) are arithmetic and calculation (*ʿilm al-ʿadad wa-l-ḥisāb*), for the materials of the Art require equilibration (*taʿdīl*), [this by means of] dissolution

107 See Newman and Grafton, "The Problematic Status."

108 Omodeo, "Perfection," p. 94. That the Copernican heliocentric doctrine was seen to be fundamentally neopythagorean is suggested by Athanasius Kircher's (d. 1680) dissimulating dismissal of Kepler's mathematical physics as a "Copernico-Pythagorean contrivance" (Rowland, "Athanasius Kircher," p. 192).

109 The Ḡābirian 'science of the balance' (*ʿilm al-mizān*), of course, is fundamentally lettrist in approach: the most accurate balance for analyzing existents is that of the letters (*mizān al-ḥurūf*) (i.e., the letters constituting the names of objects; see Melvin-Koushki, "The Quest," pp. 180–182).

(*taḥṣīl*) and synthesis (*tarkīb*) [i.e., *solve et coagula*], inasmuch as they comprise parts that are wet, parts that are dry, parts that are hot and parts that are cold; to know these [parts] one must therefore know all their precise quantities (*kammiyyāt*), and likewise all their qualities (*kayfiyyāt*), in order to realize the fundamental and equilibrating proportions and laws (*al-nisab wa-l-qawānīn al-aṣliyya tumma l-taʿdiliyya*) [that govern these materials]. I have treated this topic at length in my book *The Proof for What Pertains to the Secrets of the Balance* (*al-Burhān fī mā yataʿallaq bi-asrār al-mīzān*),¹¹⁰ wherein I indicate what of natural and metaphysical science (*al-ʿilm al-ṭabīʿī wa-l-ilāhī*) is a prerequisite [for the Art], as well as the extent to which mathematical science (*al-ʿilm al-riyāḍī*) encompasses the truths and principles of this science ...

I further hold it necessary that you know that the alchemical realm (*al-ʿālam al-ṣināʿī*) presupposes a proficiency in geometry (*ʿilm al-handasa*), this after calculation, multiplication (*ḍarb*), division (*qisma*), square root extraction (*iḥrāğ al-ğidr*), algebra (*al-ğabr wa-l-muqābala*) and ratios (*nisab al-ağzāʾ*). So too does the alchemical realm require knowledge of equal and non-equal lines (*al-ḥuṭūṭ al-mutasāwiya wa-l-ğayr mutasāwiya*), circles and the [various geometrical] shapes (*aṣkāl*), such that [the alchemist] may refine the workings of implements (*ālāt*), ovens (*tanānīr*) and [various] structures (*bunyān*) and accurately apply the principles and standard rules of thumb of geometry so as to bring to perfection all types of crafts (*ṣanāʿīʿ*) and operations. Likewise, [the alchemist must be conversant with] geodesy (*masāḥa*), area measurement (*taksīr*), finger-widths (*aṣābiʿ*) [i.e., inches], hand-widths (*qabaḍāt*) [i.e., feet], extensions (*imtidādāt*), inclinations (*inḥirāfāt*) and the secrets of circles and all other [geometrical] forms, for the Divine Art employs likenesses (*amṭāl*) and figurations (*taṣakkulāt*) in all of its operations.

As for astronomy (*ʿilm al-falak*), it is the basis of the alchemical realm (*fā-huwa l-aṣl li-l-ʿālam al-ṣināʿī*) and comprises the secrets of the sciences of the balances (*ʿulūm al-mawāzīn*) and the sciences of the keys (*ʿulūm al-mafātīḥ*) in their entirety ... As for medicine (*ʿilm al-ṭibb*), it is a branch of alchemy (*farʿ min furūʿ ʿālam al-ṣināʿa*), or rather identical to it ... Finally, the alchemical realm encompasses the secrets of the philosophical sciences (*al-ʿulūm al-falsafīyya*), as well as every other practical art. Its fruit is sweeter and more noble than that produced by any other operation, and its results sublimer and greater: it confers active control

110 More properly *al-Burhān fī asrār ʿilm al-mīzān*.

of the cosmos (*‘alay-hā madār al-‘ālam fī l-taṣarruf wa-l-af‘āl*). Therefore understand what the true philosophy (*ḥikma*) is, and dispense with absurdities!¹¹¹

Despite his status as the greatest alchemist of the century, however, al-Ğildakī is a supremely shadowy figure—the most basic details of his biography are unknown—and his influence was confined to secretive alchemical circles through the Ottoman period.¹¹² By contrast, Ibn Turka was very much a public figure, a prominent Timurid official, and upon his return to Isfahan after years of training in Cairo developed close connections with the ruling and scholarly elites of early 9th/15th-century Timurid Iran. As a member of what has been called a republic of letters,¹¹³ these relationships are traceable through collections of his surviving correspondence with and the copies of works dispatched or dedicated to various patrons and colleagues, including both Ulug Beg and Qāẓizāda Rūmī.¹¹⁴ Ibn Turka’s first and most important patron was Iskandar Sulṭān (r. 812–817/1409–1414), grandson of Temür (r. 771–807/1370–1405), ruler of Fars and contender with his uncle Šāhruḥ (r. 811–850/1409–1447) for control of Iran. While he failed in that aim, being defeated and executed by Šāhruḥ after five years of rule, Iskandar’s significance within the development of Turko-Mongol Perso-Islamic imperialism—Ottoman, Safavid, Uzbek and Mughal alike—lies in his successful performance of saint-philosopher-kingship, the first such in the post-Mongol period and hence useful model

111 MS Philadelphia LJS 441, ff. 12^b–13^a; my thanks to Nicholas Harris for this reference and his assistance with the translation. It should be noted that al-Ğildakī’s universalization of alchemy builds on similar arguments by e.g. Ibn Waḥṣiyya (d. 318/930); my thanks to Tuna Artun for this observation. On al-Ğildakī generally see Harris’s forthcoming University of Pennsylvania dissertation, the first monograph on this important but heretofore neglected figure. A similar sea-change was to occur slightly later in Europe: alchemy and astrology were considered very different and often mutually incompatible disciplines until the 15th century, when occultism was thoroughly neoplatonized and integrated by Ficino (d. 1499) and especially Agrippa, Dee and others. From this point forth they became inseparable in the projects of many, though not all, occultists; alchemy could now be called “terrestrial astrology” and astrology “celestial alchemy,” as argued by universalizing practitioners like Dee and Elias Ashmole (d. 1692) in particular. See Newman and Grafton, “The Problematic Status”; Forshaw, “Chemistry.” On the early modern interpenetration of alchemy and kabbalah see Forshaw, “*Cabala Chymica*.”

112 Harris, “A Life in the Shadows”; Artun, “Hearts.”

113 Binbaş, *Intellectual Networks*.

114 See n. 32 above. For an account of his surviving correspondence with contemporaries see Melvin-Koushki, “The Quest,” pp. 133–152.

to millenarian dynasts the Persianate world over. That performance, moreover, was largely predicated on his identification with 'Alī and the Prophetic House, on the one hand, and his patronage of astronomers, astrologers and various other occultists on the other.¹¹⁵ Their number includes in the first place Ibn Turka: his first work on lettrism, *R. Ḥurūf*, was commissioned by Iskandar, and the *Book of Inquiries* was evidently begun at his instance.¹¹⁶ Much has been made of Iskandar's interest in astronomy and astrology in particular; he employed scholars of the caliber of Maḥmūd b. Yaḥyā Kāšī, aka 'Imād al-Munaḡḡimīn, and tasked his book workshop (*kitābhāna*) with the production of gorgeous, luxury compendia on the various philosophical, mathematical, literary and religious sciences. It is not an accident that one of the only two Islamicate illuminated horoscopes to have reached us is Iskandar's.¹¹⁷ But the legitimacy of his claim to being a spiritually perfected philosopher-scientist in his own right is best seen in the preface he wrote for a comprehensive manual of mathematical astronomy (*'ilm-i hay'at*), *Ġāmi'-i sulṭānī*, possibly his own work as well. Most notably, in this preface Iskandar provides an overview of his personal intellectual and spiritual development, and therein declares astronomy-astrology (*'ilm-i nuḡūm*) to be of all sciences the most useful to kings—with the exception of lettrism (*'ilm-i ḥurūf*), the sole universal science.¹¹⁸ This remarkable assertion is inexplicable absent the context of Ibn Turka's writings and status as prominent member of Iskandar's court. Moreover, its pairing of lettrism and astrology was to prove paradigmatic: the most successful ideological platforms developed for Timurid and Aqquyunlu dynasts in the 9th/15th century and Mughal, Safavid and Ottoman dynasts in the 10th–11th/16th–17th were explicitly founded on these two sciences.

The earliest scholar to develop just such a lettrist-astrological platform in support of Timurid political legitimacy was, not coincidentally, Ibn Turka's friend, student and fellow occultist Šaraf al-Dīn Yazdī (d. 858/1454). Unlike his teacher, Yazdī devoted few works to the occult sciences as such, and achieved his fame primarily as a historian and litterateur; yet his oeuvre is permeated with occultist, and especially lettrist and astrological, sensibilities.¹¹⁹ Thus his

115 On Iskandar Sulṭān and his brilliant court in Shiraz and then Isfahan see e.g. Aubin, "Le mécénat."

116 Melvin-Koushki, "The Quest," pp. 52, 88–90.

117 Keshavarz, "The Horoscope"; Caiozzo, "The Horoscope."

118 *Dībācha-yi Ġāmi'-i sulṭānī*, in Šaraf al-Dīn 'Alī Yazdī, *Munša'āt*, pp. 207–211.

119 That lettrism was central to Yazdī's larger intellectual project is borne out in three works in particular: *Kunh al-murād fī waḥq al-a'dād*, on magic squares; *Ḥulal-i muṭarraz*, the first comprehensive formulation of the *mu'ammā* or logogriph genre, which presents it as an

celebrated history of Temür, the *Zafarnāma*, emphasizes its subject's status as the *ṣāhib-qirān*, or Lord of Conjunction, an astrological title that became thenceforth standard for all imperial claimants;¹²⁰ more notably, Yazdī proves in the preface through an analysis of Temür's horoscope that the conquerer is a historical manifestation of the *coincidentia oppositorum* or union of opposites—his Ascendant having the saturnine firmness (*ṣabāt*) of Capricorn, which is yet a cardinal or moving (*munqalib*) sign, this combination allowing for sweeping revolution (*inqilāb*) in the world—, a claim that admits of lettrist demonstration.¹²¹ (It should here be noted that the *coincidentia oppositorum* (*mağma' al-aḍḍād, mu'tanaq al-aṭrāf*) is a fundamental occultist principle appearing frequently in Ibn Turka's works, and driving, I would argue, the synthesizing mentality that pervades “early modern” Islamo-Christianate intellectual history as a whole, including the quest to harmonize Plato and Aristotle.¹²²) In the surviving *Dībāḥa* to his unfinished *Faṭḥnāma-yi ṣāhib-qirānī*, moreover, Yazdī similarly assigns to Temür a dual astrological-lettrist basis of legitimacy.¹²³ At the same time, Yazdī is responsible for valorizing geomancy as an equally useful occult art; he wrote a tract defending this science that appears to constitute a direct and incisive rebuttal of Ibn Ḥaldūn's anti-occultism arguments in the *Muqaddima*.¹²⁴ This natural association between lettrism and geomancy is further exemplified in Luṭf Allāh Nišāpūrī Samarqandī's *Hulāṣat al-*

essential skill in the lettrist's technical repertoire; and *Nikāt-i kalimāt al-tawḥīd fī l-Qur'ān*, aka *Ḥaqā'iq al-tahlīl*, a lettrist analysis of the quranic *muqatta'āt* and the statement of faith *lā ilāha illā Llāh* based on Ibn 'Arabī's *al-Futūḥāt al-makkiyya* and Ibn Turka's *K. al-Mafāḥiṣ*.

120 On this occult title see e.g. Moin, *Millennial Sovereign*, pp. 36–60.

121 *Zafarnāma*, vol. 1, pp. 9f. Or as he puts it, ‘The sweet scent of the verity of this claim may be inhaled from the truth-gardens that bloom with the secrets of the revealed isolated quranic letters (*ṣamāma-yi ṣidq-i īn da'vā az ḥadāyiq-i ḥaqāyiq-i azhār-i asrār-i muqatta'āt-i ḥurūf-i munzala-yi qur'ānī istinšāq mī-tavān nīmūd*)’ (ibid., p. 10).

122 Melvin-Koushki, “The Quest,” pp. 336, 359. The Latin term was first introduced by Nicholas of Cusa (d. 1464), Ibn Turka's later contemporary; see e.g. Martin, “The Dialectical Process.” In his *Religion after Religion* Stephen Wasserstrom offers a penetrating and wide-ranging discussion of the concept in the History of Religions movement and in the context of poetics, politics and ethics; as he shows, Gershom Scholem, Mircea Eliade and Henry Corbin were all great advocates of the principle, as was Carl Jung, who preferred the term *complexio oppositorum* (see e.g. his *Mysterium Coniunctionis*); and Freud attempted to explain the union of opposites in language and religion in his essay “The Antithetical Sense of Primal Words.”

123 Binbaş, *Intellectual Networks*, pp. 215–216. Specifically, Temür is there shown to be the messianic *āḫir al-zamān* on lettrist grounds.

124 This tract is translated and studied in Melvin-Koushki, “Defending Geomancy.”

baḥrayn, written in 812/1409, which treats of geomancy and talismanically operative magic squares (the two seas of the title) as complementary sciences: the first tells you the future, and the second allows you to change it—a prospect of great attractiveness to ruling elites.¹²⁵ From the early 9th/15th century onward, then, the three elite sciences used to legitimize and advance imperial claims in the Persianate world, and hence the occult sciences most heavily patronized, were lettrism, astrology and geomancy, due in large part to the work of Ibn Turka and Šaraf al-Dīn Yazdī as the two most prominent intellectuals of early Timurid Iran.¹²⁶ Any analysis of later Arabo-Persian astrology that neglects either lettrism or geomancy must therefore be considered abortive.

The status of lettrism and astrology as an interdependent unit only increased in salience during the course of the 9th/15th century. Ġalāl al-Dīn Davānī (d. 908/1502), for instance, ideological mainstay of the Aqquyunlu Empire, follows Yazdī in putting his patron Uzun Ḥasan's (r. 861–882/1457–1478) political legitimacy on a primarily lettrist-astrological footing in his seminal *Aḥlāq-i Ġalālī* and elsewhere.¹²⁷ Davānī's model was further developed by other historians working at the Aqquyunlu court, including in the first place his students Faẓl Allāh Ḥuṅgī Iṣfahānī (d. 927/1521) and Idrīs Bidlīsī (d. 926/1520); the latter then repurposed it for Ottoman use in his *Hašt bihišt*.¹²⁸ That lettrist-astrological platforms for political legitimation remained in vogue in the Persianate world until at least the mid-11th/17th century—and in some cases down to the 14th/20th¹²⁹—is confirmed by the presence of just such an argument in the preface to the *Kārnāma-yi šāhib-qirān-i šānī*, a comprehensive work on astronomy commissioned by Šāhḡahān (r. 1037–1068/1628–57), which is concerned to demonstrate the ontological equivalency of the Mughal emperor with Temür, and of both rulers with the astrological title Lord of Conjunction, by means of lettrist analysis.¹³⁰ (In view of the great and continued success of

125 See Melvin-Koushki, "Defending Geomancy." The two historians possibly met in Cairo (Ibn Ḥaldūn lived in that city from 785/1383 to his death in 808/1406, and Yazdī spent at least part of the period 795–810/1393–1408 there with Ibn Turka), and certainly moved in the same court circles while there.

126 See e.g. Dawlatšāh Samarqandī (d. 900/1494 or 913/1507), *Tazkirat al-šu'arā*, p. 340.

127 See Melvin-Koushki, *The Occult Science*.

128 Melvin-Koushki, "The Quest," p. 308.

129 As with, e.g., Nādir Šāh Afšār (r. 1736–1747), Faṭḥ 'Alī Šāh Qāḡār (r. 1797–1824), Naṣr Allāh Ḥan Mangīt (r. 1827–1860), etc.

130 Moin, *Millennial Sovereign*, pp. 23 f. Šāhḡahān, of course, is likewise responsible for the Taj Mahal, a masterpiece of Timurid architecture—one that functions as a giant talisman, the ground plan being a magic square (my thanks to Julia Bray for this observation).

Yazdī's original model, in fact, I propose that the title *ṣāhib-qirān* (aka Starlord) be itself conjuncted with another: *ṣāhib-ḥurūf*, Letterlord.¹³¹) Davānī was also the author of a popular introductory manual of astrological letter magic, *Tuḥfa-yi rūḥānī fi ḥavāṣṣ al-ḥurūf*, written for a Khalji sultan of Malwa in central India, that quotes from Ps.-Aristotle's *Book of Conjunctions* (*K. Iṣṭikākāt*) and cites Ibn Turka's lettrist prognostication for Šāhruḥ of the duration of the Timurid state.¹³² More importantly for our purposes here, in his *R. Tahlīliyya* Davānī appears to uphold Ibn Turka's—and Iskandar Sulṭān's—ranking of lettrism as queen of the sciences, epistemologically superior to, in ascending order, scriptural literalism, logic and sufi theory.¹³³

The various occult sciences, now firmly rooted in number theory thanks to Ibn Turka and his neopythagoreanizing brethren—who, significantly, refer to themselves as *Iḥwān al-Ṣafā'*—,¹³⁴ continued to grow in popularity as mainstream intellectual pursuits and indispensable props to political legitimacy throughout the Persianate world during the 10th/16th century. Here again Persian classifications of the sciences serve as an efficient barometer of this trend. For reasons of space four examples from the 10th/16th and 11th/17th centuries must suffice, the first two Safavid and the second two Mughal.

The *Sullam al-samāwāt* of Abū l-Qāsim Anṣārī Kāzīrūnī (b. 966/1559), a prominent Shirazi scholar under Shah 'Abbās (r. 995–1038/1587–1629) and student of Afzal al-Dīn b. Ḥabīb Allāh Turka (d. 991/1583) (presumably the great-great-great-grandson of Ibn Turka), is an eclectic, genre-bending work equal parts theological tract, history of philosophy, literary history and encyclopedia of the sciences. Crucially, in the *Sullam al-samāwāt*, completed by 1014/1605, Kāzīrūnī serves Safavid imperial needs admirably by constructing an intellectual-religious pedigree for his patrons that is simultaneously Twelver Shi'i,

131 While the term *ṣāhib-ḥurūf* does not occur as such in the sources, I propose it here as a heuristic for flagging the central importance of lettrist arguments to imperial ideology construction from the Timurids onward.

132 See Melvin-Koushki, "The Quest," pp. 247–255. This manual is edited, translated and contextualized in Melvin-Koushki, *The Occult Science*.

133 The *R. Tahlīliyya* (ed. Firīšta Furūzanda) appears to be inspired in the main by Šaraf al-Dīn Yazdī's similarly lettrist *Ḥaqā'iq al-tahlīl*; for a summary of the former see Melvin-Koushki, "The Quest," pp. 256–261. Cf. Davānī's scheme in his *Naqd-i niyāzī*, which interprets a *dūbayt* and a ghazal by Ḥāfiẓ from three perspectives: that of the *muvaḥḥida*, that of the *ṣūfiyya* and that of the *ḥukamā-yi maššā'īyya u iṣrāqīyya* (p. 44). Given the above context, it is significant that Davānī explicitly distinguishes between the *ahl-i tawḥīd*—i.e., lettrists—on the one hand and the sufis and the peripatetic-illuminationist philosophers on the other in a manner similar to the *Dibācha-yi Ğāmi'-i sulṭānī* of Iskandar Sulṭān.

134 Melvin-Koushki, "The Quest," pp. 16–19, 54f.

sufi, occultist and neopythagorean-neoplatonic: an impressive feat of synthesis indeed, and one reflective of Ṭahmāsb's (r. 930–983/1524–1576) and 'Abbās's experimentalist efforts to create a new, more stable Safavid imperial culture. Two chapters of seven, or roughly a quarter of the text, are formally a biographical dictionary of philosophers and of poets respectively; the philosophers covered run from Fārābī and Ibn Sīnā to Davānī and Ġiyāṣ al-Dīn Daštakī (d. 949/1542) by way of Faḥr al-Dīn Rāzī, Ṭūsī, Quṭb al-Dīn Šīrāzī, the Iḥwān al-Šafā', Ibn 'Arabī and others. Brandishing his neopythagorean loyalties, Kāzīrūnī declares of the Iḥwān al-Šafā', for instance:

There is not a single word in the *Epistles* [of the Brethren of Purity] that this wretch has not read repeatedly with the eye of desire and longing ... In some of these epistles it is stated that on most topics their discussion accords with the doctrine of Pythagoras.¹³⁵

He also lionizes Ibn 'Arabī as a universalist philosopher, his *al-Futūḥāt al-makkiyya* combining traditionalist (*naqlī*), rationalist (*'aqlī*), inspired (*kashfī*) and intuitive (*ḡawqī*) modes of discourse.¹³⁶ Closer to his own day, while Kāzīrūnī declines to take sides in the infamous Daštakī-Davānī controversy, apart from noting that philosophers of his own day compared their rivalry to that between Fārābī and Ibn Sīnā or Plato and Aristotle, he does emphasize the old-fashioned and outmoded nature of Ṣadr al-Dīn Daštakī's (d. 903/1498) approach—thereby hinting at the high Safavid modishness of Davānī's neopythagoreanism.¹³⁷

The most important chapter for our purposes here, however, is that dedicated to a taxonomy of the occult sciences as a subset not of natural or even mathematical philosophy but of *walāya*, here presented as sufi-style sainthood rather than a Twelver theological category. Simply put, it is here that the sanctification of occultism, a process that began in the 7th/13th century with Ibn 'Arabī and al-Būnī and was consolidated by Ibn Turka and al-Biṣṭāmī in the 9th/15th, is finally and formally canonized. After treating of sainthood (*walāyat*), saintly miracle-working (*karāmat*), saintly inspiration (*mukāṣafat*)

135 *Sullam al-samāwāt*, p. 205. Kāzīrūnī believes the *Epistles* to be the work of a single author who died sometime between 500–550/1107–1155—i.e., that they are a firmly 6th/12th-century work—and implies that he consulted them in their abridged Persian translation entitled *Muḡmal al-ḥikma* (pp. 205, 207).

136 *Sullam al-samāwāt*, p. 208.

137 *Sullam al-samāwāt*, pp. 220–223; on the Daštakīs as committed Avicennans see Pourjavady, *Philosophy*, pp. 16–24; Bdaiwī, "Shī'ī Defenders."

and other miraculous feats (*ḥāriq-i ʿādāt*), Kāzirūnī proceeds to discuss a number of occult sciences in the following sequence: alchemy (*kīmīyā u iksīr*), talismans (*līmīyā*), letter magic (*sīmīyā*), prestidigitation and illusionism (*rīmīyā u šuʿbadāt*), terrestrial magic (*hīmīyā*), letter divination (*ǧafr*), talismanic magic squares (*aʿdād*), divine-names magic (*ḥavāṣṣ-i asmāʾ-i Allāh*), astronomy-astrology (*nuǧūm*) and quranic magic (*ḥavāṣṣ-i suvar u āyāt*), ending with general observations and a selection of magically effective prayers or incantations (*adʿiya-yi muǧarraba*).¹³⁸ Interestingly, in the section on *ǧafr* (here used metonymically to refer to *ʿilm-i ḥurūf* as a whole) Kāzirūnī appears to cite Ibn Ḥaldūn's classification of this science as a subset of *sīmīyā*, rather than reiterating Āmulī's formal separation of the two.¹³⁹ In the section on divine-names magic, moreover, he identifies al-Būnī as the science's foremost practitioner in Islam.¹⁴⁰ As for *ʿilm-i nuǧūm*, Kāzirūnī categorizes it as being only partially an occult science (*az ʿulūm-i ǧarība*)—that is, formally distinguishing between astronomy and astrology in purely occultist terms, not natural-philosophical. Judicial astrology (*aḥkām-i nuǧūm*), he states, constitutes a rarer (*nādir*) subset of the science of the stars, and is focused on such matters as conjunctions (*qirānāt*) and horoscopes (*ṭālīʿ-i mawlūd*).¹⁴¹

In contrast to Kāzirūnī's eclectic work, the *Riyāz al-Abrār* of Ḥusayn ʿAqīlī Rustamdārī, completed in the Safavid capital, Qazvin, in 979/1571 toward the end of Shah Ṭahmāsb's reign, is a formal Persian encyclopedia clearly descended from the *Ǧāmiʿ al-ʿulūm* and the *Nafāyis al-funūn*, treating of approximately 90 sciences in systematic fashion (hence its alternate title *Kitāb-i Tisʿīn*).¹⁴² Yet its classification scheme is highly idiosyncratic, particularly with respect to the occult sciences—to which Rustamdārī, like Kāzirūnī, devotes a significant percentage of the work. Moreover, Rustamdārī seeks to "Twelverize" his encyclopedia to an even greater extent than Kāzirūnī does the *Sullam al-samāwāt*. To this end, he opens with a long prefatory section arguing for Imam

138 *Sullam al-samāwāt*, pp. 81–130. *Kīmīyāʾ*, *līmīyāʾ*, *hīmīyāʾ*, *sīmīyāʾ* and *rīmīyāʾ* often occur in a series, their initial letters being combined to produce the occultist motto 'the whole is a secret' (*KLHSR* = *kullu-hu sirr*). The definitions of these terms are not entirely stable from author to author; *sīmīyāʾ*, for instance, can refer to anything from letter magic or divine-names magic to illusionism. Cf. Ḥusayn Vāʾiẓ Kāšifī's *Asrār-i Qāsimī*, an occultist manual of considerable popularity in the Safavid period, which opens by defining these five sciences somewhat differently as alchemy, talismans, planetary magic (*tashḥirāt*), illusionism (*ḥayālāt*) and terrestrial magic (*šuʿbadāt*) respectively (pp. 3f.).

139 *Sullam al-samāwāt*, p. 119.

140 *Sullam al-samāwāt*, p. 120.

141 *Sullam al-samāwāt*, pp. 121f.

142 Storey, *Persian Literature*, vol. 2, pt. 3, p. 359, no. 597.

‘Alī as the source of all sciences and offering an exegesis of his famous Šaqṣaqī sermon, then uses the first chapter, on religions and sects, to demonstrate that the ‘saved sect’ (*al-firqa al-nāḡiya*) referred to in the Prophetic hadith can only be the Twelver Shi‘a. Other Twelver features are interspersed throughout the *Rīyāz al-abrār*, including a discussion of the apocalyptic events surrounding the parousia of the twelfth Imam (*ṣāhib al-zamān*); more importantly for our purposes, it presents lettrism as a science peculiar to the Imams, thereby establishing its derivation from *walāya*.

The *Rīyāz al-abrār* is divided into twelve chapters, or *rawzas*, one of which, the ninth, is almost entirely given over to various occult sciences; other occult sciences are classified elsewhere. Thus in the fifth *rawza*, Rustamdārī assigns oneiromancy (*ta‘bīr-i ḥ‘āb*) to the same category as ethics, economics and medicine. He devotes the sixth *rawza* to various disciplines depending on calculation (*‘ilm-i ḥisāb*) and the construction of ephemerides (*taqvīmāt*), including star tables (*zīḡāt*), judicial astrology (*aḥkām-i nuḡūmī*), horoscopes (*tālī‘-i mawlūd*) and astrolabes and other astronomical instruments (*ālāt-i aṣṭurlāb*), then discusses both geomancy (*aḥkām-i raml*) and scapulomancy (*‘ilm-i šāna*) directly after a section on eschatology and the *Ġāmāsbnāma*. The seventh *rawza* treats of physiognomy (*fīrāsāt*) together with gemmology (*ḡavāḥir*), agriculture (*filāḡat*), mechanics (*ḡarr-i aṣḡāl*), veterinary medicine (*bayṭara*), falconry (*buzāt*), pharmacology (*‘ilm-i ṣaydana*) and other applied natural sciences. Finally, the ninth *rawza* covers the remaining occult sciences, together with two sciences only semi-occult at best, as follows:

Meteorological phenomena (*kāyināt-i ḡaw*); letter divination, aka the comprehensive prognosticon (*ḡafr-i ḡāmī‘, ḡābiya*); planetary magic (*‘ilm-i taṣḡīrāt*); jinn magic (*‘ilm-i ‘azāyim*); celestial magic or talismans (*‘ilm-i ṭīlismāt*); terrestrial magic (*‘ilm-i nīranḡāt*); the yogic sciences of breath control and visualization (*‘ilm-i dam u vaḡm*); soul projection (?) (*‘ilm-i taṣarruf-i naḡs u muḡayyabāt*); alchemy, *ḡīmīyā*, letter magic and illusionism as a cohesive series (*‘ilm-i kīmīyā u ḡīmīyā u sīmīyā u rīmīyā*);¹⁴³ and magic squares (*‘ilm-i a‘dād*) (here somewhat bizarrely linked with war machines (*‘ilm-i ālāt-i ḡurūb*)).¹⁴⁴

That is to say, Rustamdārī appears to classify both astrology and geomancy as applied mathematical sciences, while at the same time decoupling them from the quadrivium (including astronomy (*‘ilm-i ḡay‘āt*)) and the sciences imme-

143 These terms are unstable; see n. 138 above.

144 War machines are normally categorized under the rubric of *ḡiyāl*, suggesting that the manuscript(s) in question may be defective.

diately derived therefrom, including algebra, geodesy and optics, all of which are given their own *rawza*, the tenth. He also separates out sciences like letter magic, letter divination and alchemy from both the mathematical and the natural sciences, much in the manner of Kāzīrūnī: this implies a *sui generis* status in their association with the sacral power of the Imams. That Rustamdārī does not consider astrology and geomancy (or oneiromancy and physiognomy) to be of an epistemological unit with the other occult sciences is less explicable, however, and suggests that a closer study of this unique encyclopedia, as yet unpublished, is in order.¹⁴⁵

This tendency to both mathematicalize and sanctify the occult sciences was hardly exclusive to Persian encyclopedists working in the Safavid realm: Mughal India was equally implicated. Perhaps the best example here is the *Ġavāhir al-ʿulūm-i humāyūnī* of Muḥammad Fāzīl Miskīnī Qāzī Samarqandī, written for Emperor Humāyūn (r. 937–963/1530–1556) (here addressed as Muḥammad Humāyūn *pādīshāh al-ġāzī*) shortly after the latter's return to power in 962/1555.¹⁴⁶ While it too is explicitly modeled in the main on Āmulī's *Nafāyis* and Rāzī's *Ġāmiʿ*, a fact noted in the work's preface, its approach to the occult sciences effectively combines those of Kāzīrūnī and Rustamdārī (minus, of course, the Twelver branding). It is also impressively occultist: almost a full sixth of the text is devoted to the 'strange sciences'—no longer particularly strange, to say the least.

The *Ġavāhir al-ʿulūm* is divided into three *maqālas*, each of which is subdivided into two *qisms*. The first two *maqālas* treat of the various literary, religious and political sciences, as well as natural sciences like medicine and hunting; the third *maqāla*, however, is largely given over to sufism and the magical properties of the Quran on the one hand and the mathematical sciences on the other, over half of which also happen to be occult. (The work's epilogue, appropriately, is devoted to eschatology (*ʿalāmāt-i āḥir al-zamān*).) Its contents are as follows:

Part 1: tafsir (*ʿilm-i tafsīr-i Qurʾān u ḥall-i alfāz-i muškila-yi qurʾānī*); the seven quranic readings, in verse (*ʿilm-i qirāʾat-i sabʿa, bi naẓm*); various

145 I did not have access to the relevant MSS at the time of writing (extant copies are at the Āstān-i Quds Library in Mashhad and the British Library), so rely here on the summary given by ʿAbd Allāh Nūrānī in the introduction to his edition of Kāzīrūnī's *Sullam al-samāwāt*, pp. xlix–liii.

146 Or, according to Muḥammad Ḥusayn Tasbiḥī, written on the occasion of Humāyūn's accession to the throne in 937/1530, probably in Samarkand, though a copy of the work may have never made it to India (*“Ġavāhir”*). On this work see Brentjes, “The Mathematical Sciences,” p. 17; Storey, *Persian Literature*, vol. 2, pt. 3, pp. 358f, no. 596.

magical prayers of protection, the magical properties of quranic suras and verses and a verse translation of al-Buṣīrī's *Ode of the Mantle* (*'ilm-i ḥavāṣṣ-i awrād-i fathīyya u tarjuma-yi qaṣīda-yi burda bi naẓm u ḥizb al-baḥr u suvar u āyāt*); traditional prayers and incantations (*'ilm-i ad'ya-yi ma'sūra u da'avāt-i mashhūra*); hadith (*'ilm-i ḥadīs*); hadith criticism (*'ilm-i uṣūl-i ḥadīs*); sufi theory and terminology (*ma'rifat-i qavā'id u iṣtilāḥāt-i ṣūfiyya*); the sufi path (*'ilm-i sulūk*); the science of unification and the levels of unveiling (*'ilm-i tawḥīd u marātib-i mukāṣafāt*); sufi witnessing (*ma'rifat-i muṣāhadāt*), the sufi stations and their degrees (*ma'rifat-i maqāmāt u marātib-i ān*), the science of the real (*'ilm-i ḥaqīqat*).

Part 2: solar and lunar ephemerides and elections (*ma'rifat-i taqvīm-i šamsī u qamarī u iḥtiyār-i sā'āt*); letter magic (*'ilm-i sīmīyā*); divine-names magic and its conditions (*'ilm-i da'vat-i asmā' u šarāyit-i ān*); the construction of ephemerides and star tables (*ma'rifat-i istiḥrāğ-i taqvīm u šabaka-yi nuğūmī*); judicial astrology (*ma'rifat-i aḥkām-i nuğūm*); astronomy (*'ilm-i hay'at*); the astrolabe (*'ilm-i aṣṭurlāb*); the celestial spheres (*ma'rifat-i kurra-yi aflāk*); the seven climes (*ma'rifat-i aqālīm-i sab'a*); the constellations (*'ilm-i šuvar-i kavākib*); administrative geography (*ma'rifat-i masālik u mamālik-i ālam*); fraction of numeration (*'ilm-i taksīr*);¹⁴⁷ magic squares (*'ilm-i a'dād-i vaḥq*); the science of letters (*'ilm-i ḥurūf*); letter divination, aka the comprehensive prognosticon (*'ilm-i ḡafr-i ḡāmī'*); celestial and terrestrial magic (*'ilm-i ṭilismāt u nūranğāt*); alchemy (*'ilm-i kīmīyā*); planetary magic (*'ilm-i tashīr-i kavākib*); jinn magic (*'ilm-i 'azāyim*); geomancy (*'ilm-i raml*); calculation (*'ilm-i ḥisāb*); geodesy, mechanics and visible objects(?) (*'ilm-i masāḥat u ḡarr-i aṣqāl u bayān-i muḃṣarāt*); comptrolling (*'ilm-i istīfā*); physiognomy (*'ilm-i qiyāfa*); oneiromancy (*'ilm-i ta'bīr-i ḥ'wāb*); spasmatomancy, scapulomancy, determining the horoscope of a question, augury and ornithomancy, etc. (*ma'rifat-i iḥtilāğāt u 'ilm-i šāna u ṭālī'-i mas'ala u ma'rifat-i tafa'ul u taṭayyur u amṣāl-i ān*); horoscopes (*ma'rifat-i ṭālī'-i mavālid u zāyīca-yi ṭālī'*); Euclidean propositions (*ma'rifat-i aṣkāl-i Uqlīdis*); the Middle Books (*'ilm-i mutavas-sitāt*); music (*'ilm-i mūsīqī*); yogic visualization (*'ilm-i vahm*); chess (*'ilm-i šaṭranğ*).¹⁴⁸

147 The term *taksīr*, here clearly referring to the well-known lettrist technique, can also refer to the science of area measurement, as with al-Ğildakī above.

148 Here again I rely on the summary given by 'Abd Allāh Nūrānī in the introduction to his edition of Kāzīrūnī's *Sullam al-samāvāt*, pp. xlvii–xlix, on the basis of a manuscript copy in the Ganj Bakhsh Library; the encyclopedia is unpublished.

Miskīnī's classification schema, it is almost gratuitous to note, is striking in a) its heavy emphasis on lettrism and subsidiary lettrist disciplines, and b) its implicit, unprecedented mathematicalization even of sciences like alchemy, physiognomy and the full range of mantic techniques, from oneiromancy to spasmatomancy, never before so closely associated with the quadrivium in the Arabo-Persian encyclopedic tradition. (One is reminded here of al-Ğildakī's own mathematicalization of alchemy.) It further posits the same epistemological continuity between sufi-style *walāya* and occultism so blatantly asserted by Kāzīrūnī. In sum, the *Ğavāhir al-‘ulūm-i humāyūnī* epitomizes this article's main thesis: that many occult sciences, including astrology, lettrism and geomancy, were progressively and simultaneously mathematicalized and sanctified, a process that climaxed in the 10th/16th-century Persianate east.

The status of astrology and geomancy in particular as exclusively mathematical sciences is further confirmed by contemporary Mughal manuals of occultism, including the *Qavā'id al-hidāya*,¹⁴⁹ on geomancy, written by Hidāyat Allāh Munağğim-i Šīrāzī (fl. 962–1001/1555–1593) for Emperor Akbar in the month of Farvardīn in year 37 of Akbar's Ilāhī calendar—that is, the spring equinox of the year 1001 (March 1593): the moment of dawning of the Islamic millennium over the horizon of Akbar's sacred body, itself effulging forth the talismanic cosmocrator's eschatological imperium over all of India. The *Qavā'id al-hidāya* was thus one of the many works the Mughal emperor commissioned to celebrate the millennium's arrival, including the feted *Ā'in-i Akbarī* and the *Tārīḥ-i alfi*. The prologue of the work, not incidentally, is entirely astrological in its symbolism, a design feature sure to catch the fancy of Akbar, a great star worshipper and card-carrying member of the so-called *ahl-i nuğūm u tanāsuh*.¹⁵⁰ Hidāyat Allāh classifies both geomancy and astrology as follows:

Know (God guide you) that philosophy (*ḥikmat*) is divided into three categories: the natural sciences (*‘ilm-i ṭabī‘ī*), the mathematical sciences (*‘ilm-i rīyāzī*) and metaphysics (*‘ilm-i ilāhī*). The principal mathematical sciences are four: astronomy (*ḥay'at*), arithmetic ([*‘adad*]),¹⁵¹ geometry

149 See Melvin-Koushki, "Persianate Geomancy"; Storey, *Persian Literature*, vol. 2, pt. 3, p. 481, no. 850 (2, 4, 5); Āqā Buzurg, *al-Darī'a*, vol. 17, p. 195, no. 1034.

150 Akbar's preoccupation with planetary symbolism and the doctrine of metempsychosis are both well-known, and seem to have been inspired in part by Āzar Kayvānī immigrants to India; see Sheffield, "Lord of the Planetary Court." Muḥammad Dihdār Šīrāzī (d. 1016/1607), Maḥmūd Dihdār's son, speaks of an *ahl-i nuğūm u tanāsuh* clearly in reference to this group ("Durr al-Yatīm," in *Rasā'il-i Dihdār*, p. 131).

151 The ms I consulted has الحية, clearly a scribal error.

(*handasa*) and music (*mūsīqī*). The mathematical disciplines derived from these are numerous, and include geomancy (*raml*) and astrology (*nuḡūm*), which are exceedingly noble and useful sciences.¹⁵²

The prestige that these two mathematicalized occult sciences enjoyed at the Mughal court and among persophone Indian elites more generally is indicated by no less an authority than Abū l-Faẓl ‘Allāmī (d. 1011/1602), primary architect of the new Mughal imperial culture, who in the *Ā’in-i Akbarī* explicitly lists them both as part of his ideal core curriculum for all students in Akbar’s realm, to be studied even before medicine or logic:

The following sciences should be acquired consecutively: ethics, arithmetic, accounts, agriculture, geodesy, geometry, astronomy-astrology (*nuḡūm*), geomancy (*raml*), economics, government, medicine and logic, then [the rest of] the natural and mathematical sciences and metaphysics, then history.¹⁵³

Moreover, the increasingly close association between astrology and geomancy is reflected in a terminological shift in later Persian usage: from the 9th/15th century, geomancers (sg. *rammāl*) could also be designated by the term *munaḡḡīm*, suggesting the thorough blending of the offices of astrologer and geomancer.¹⁵⁴ An example of this phenomenon is Hidāyat Allāh himself, a Shirāzi *munaḡḡīm*, who is only known to have produced a work on geomancy for his royal patron.

That this new schema continued hale and hearty into at least the 11th/17th century is confirmed by another Mughal encyclopedia, the *‘Uqūl-i ‘ašara*, completed in 1084/1673 by the otherwise unknown Muḥammad Barārī Qāqšāl, *taḥalluṣ* Ummī.¹⁵⁵ Its value for our purposes here lies precisely in the fact that it is not at all cutting-edge, apart from its novel organizational schema: the *‘Uqūl-i ‘ašara* reflects the scientific consensus established well over a century prior. The work is divided into ten ‘intellects’ (sg. *‘aql*), which are progressively subdivided into ‘understandings’ (sg. *fahm*), ‘discernments’ (sg. *firāsāt*) and ‘sagacities’ (sg. *kiyāsāt*); the first three intellects treat of sciences pertaining to

152 *Qavā’id al-hidāya*, MS Majlis 12563, p. 312. See also Brentjes, “The Mathematical Sciences.”

153 *Ā’in-i Akbarī*, vol. 1, p. 202.

154 Melvin-Koushki, “Persianate Geomancy.”

155 The *‘Uqūl-i ‘ašara* was summarized and its manuscript copies identified by Īraḡ Afšār (“*‘Uqūl-i ‘ašara*”); see also Storey, *Persian Literature*, vol. 2, pt. 3, pp. 359 f., no. 598; Brentjes, “The Mathematical Sciences,” p. 18.

the celestial realm (*kurra-yi aflāk*), the last seven of sciences pertaining to the terrestrial realm (*kurra-yi zamīn*). Unsurprisingly, the three overarching categories deemed most representative of the celestial-mathematical disciplines are 1) astronomy-astrology, 2) the astrolabe and other observational instruments and 3) geomancy. For their part, talismanic magic (*ṭilism*) and prophetic and saintly miracles (*mu'ǧiza u karāmāt*) are assigned to the fourth intellect, or the first category of terrestrial sciences, and physiognomy (*qiyāfa*) to the fifth intellect; there appears to be no discussion of other occult sciences like alchemy or the various mantic disciplines. This scientific consensus is further reflected by a second encyclopedia from the same mid-11th/17th-century Mughal context, the *Šāhid-i šādiq* of Muḥammad Šādiq b. Muḥammad Šālih Iṣfahānī Āzādānī (d. 1061/1652), which is even more devoted to the astronomical-cosmographical sciences, and similarly classifies astrology strictly as an applied mathematical science.¹⁵⁶

But what of the Ottomans? The Ottoman Empire, of course, was an intrinsic part of the Persianate world in many important respects, this despite its self-positioning as heir of Byzantium and conflict with the Safavid state; more pertinently, high Ottoman imperial culture was at least as, if not more, occultist than its Safavid and Mughal cognates during the 10th/16th century, as has been definitively shown by Cornell Fleischer, İhsan Fazlıoğlu and others.¹⁵⁷ Yet Ottoman Arabic encyclopedism appears to fundamentally diverge from the Persian tradition from the 8th/14th century onward, especially with respect to the classification of the occult sciences: simply put, they are not mathematicalized, nor are they sanctified.

While space does not permit a survey of Ottoman encyclopedias here, my preliminary analysis suggests that they, unlike Persian encyclopedias, remain wedded to the Avicennan model as formulated in the first place by Ibn al-Akfānī (d. 749/1348) in his *Iršād al-qāsid ilā asnā l-maqāsid*.¹⁵⁸ Written in Mamluk Cairo by a practicing physician, the *Iršād al-Qāsid* served as a model for subsequent Mamluk and Ottoman encyclopedists of the caliber of al-Qalqašandī and Taşköprüzāde (d. 968/1561). Ibn al-Akfānī, as a good Avicennan, naturally classifies all occult sciences as derivative natural sciences, which include:

156 Brentjes, "The Mathematical Sciences," pp. 17 f.

157 See e.g. Fleischer, "Ancient Wisdom"; idem, "Shadow"; Fazlıoğlu, "İlk Dönem"; Gril, "Ésotericism"; idem, "L'énigme." The same has been shown by e.g. Kathryn Babayan (*Mystics*) and Azfar Moin (*Millennial Sovereign*) for the Safavid and Mughal contexts respectively.

158 See Witkam, "Ibn al-Akfānī"; *Iršād al-qāsid*, ed. Jan Just Witkam.

Medicine (*‘ilm al-ṭibb*), veterinary medicine and falconry (*‘ilm al-bayṭara wa-l-bayzara*), physiognomy (*‘ilm al-firāsa*), oneiromancy (*‘ilm ta’bīr al-ru’yā*), judicial astrology (*‘ilm aḥkām al-nuḡūm*), magic (*‘ilm al-siḥr*), talismans (*‘ilm al-ṭilasmāt*), illusionism (*‘ilm al-simiyā’*),¹⁵⁹ alchemy (*‘ilm al-kīmīyā’*), agriculture (*‘ilm al-filāḥa*).

As he clarifies, these natural sciences treat of either simple bodies or composite bodies or both. Thus astrology treats of celestial simple bodies, and talismans treat of terrestrial or elemental simple bodies. As for composite bodies, illusionism treats of those that have no constitution (*mizāj*), while those that have constitutions are further subdivided into those that do not have souls (alchemy) and those that do have souls; here there are either insensible (*ḡayr mudrika*) souls (agriculture) or sensible (veterinary medicine for animals, medicine for humans). Physiognomy then inquires into the inner states of rational human souls while they are conscious, and oneiromancy does the same while they are unconscious. That science which treats of both simple and composite bodies is magic.¹⁶⁰ Unlike Ibn Ḥaldūn, then, a near-contemporary fellow Cairene, Ibn al-Akfānī is more than happy to admit the epistemological legitimacy of the occult sciences; indeed, for him they are crucial components of the edifice of natural philosophy (and psychology) as a whole.

An even stronger pro-occultist position is taken by Taşköprüzāde in his comprehensive *Miftāḥ al-sa’āda wa-miṣbāḥ al-siyāda*.¹⁶¹ Completed in 948/1541, it represents a culmination of the Arabic encyclopedic tradition, and served as the primary model for subsequent Ottoman encyclopedias, including most prominently the *Kaşf al-ẓunūn* of Ḥāḡḡī Ḥalīfa (d. 1067/1657).¹⁶² Significantly for our purposes in this article, Taşköprüzāde was a student of Mirim Çelebi (d. 931/1525), the great-grandson of both Qāzizāda Rūmī and ‘Alī Qūščī, who broke with his radically mathematicalizing forebears to reintroduce aristotelian physical principles into his astronomy;¹⁶³ Taşköprüzāde is therefore well-

159 Ibn al-Akfānī, oddly enough, here conflates *simiyā’* with *ša’baḍa*; several decades later, Ibn Ḥaldūn clearly applies the term to letter magic.

160 *Iršād al-qāṣid*, p. 46 (Arabic text).

161 On this work see Bellino, “Classification.”

162 Unlike Taşköprüzāde, however, Ḥāḡḡī Ḥalīfa takes a somewhat ambiguous position on the occult sciences, particularly in his discussions of astrology and alchemy (*Kaşf al-ẓunūn*, vol. 2, cols. 1527–1533, 1930–1931).

163 Fazlhoğlu, “Between Reality and Mentality,” p. 35. On Mirim Çelebi, and particularly his contributions to astrology, see Ahmet Tunç Şen’s forthcoming University of Chicago dissertation.

versed in the Timurid-Ottoman astronomical tradition, and also draws on thinkers like Mullā Fanārī (d. 834/1431), a neoplatonizing member of the Iḥwān al-Ṣafā' network.¹⁶⁴

The *Miftāḥ al-sa'āda* is organized into seven branching 'trees' (sg. *dawḥa*), the first of which is devoted to the sciences of writing—an ordering certainly inspired by Ibn Turka's unprecedented privileging of written language over spoken—and the fourth to the metaphysical, mathematical and natural sciences. The main divisions of this fourth section are directly modeled on Ibn al-Akfānī's treatment of the same, and in some cases copy whole subsections verbatim, but considerably expand on it to include the branches of branches (*furū' al-furū'*) of the natural and mathematical sciences—this especially in the case of the occult sciences. Thus Taşköprüzāde includes elections (*'ilm al-iḥtiyārāt*), geomancy (*'ilm al-raml*), bibliomancy (*'ilm al-fa'l*), lots (*'ilm al-qur'a*) and ornithmancy and augury (*'ilm al-ṭira wa-l-zağr*) as branches of judicial astrology, itself a branch of natural philosophy; and he massively enlarges on Ibn al-Akfānī's discussion of magic to include 14 sub-branches thereof as follows:

Soothsaying (*'ilm al-kihāna*), terrestrial magic (*'ilm al-nūranġāt*), active properties (*'ilm al-ḥawāṣṣ*), incantations (*'ilm al-ruqā*), spirit magic (*'ilm al-ʿazā'im*), jinn magic (*'ilm al-istiḥdār*), planetary invocations (*'ilm daʿwat al-kawākib*), magical diagrams (*'ilm al-falaqtīrāt*), concealment (*'ilm al-iḥfā'*), Sasanian wealth magic (*'ilm al-ḥiyal al-sāsāniyya*), jewel magic (*'ilm kashf al-dakk wa-ḥdāḥ al-shakk*), illusionism (*'ilm al-sha'baḍa wa-l-taḥayyulāt*), deception magic (*'ilm ta'alluq al-qalb*), medical magic (*'ilm al-isti'āna bi-ḥawāṣṣ al-adwiya wa-l-mufradāt*).¹⁶⁵

164 Melvin-Koushki, "The Quest," pp. 64, 67; see Shiraz Sheikh's forthcoming University of Toronto dissertation, "Molla Fenari and the Akbarian Tradition in the Early Ottoman State." On the astronomical interests of the Fanārī circle see Ragep, "Astronomy," which discusses the *Unmūdağ al-ʿulūm* of Mullā Fanārī's son Muḥammad Ṣāḥ Fanārī (d. 839/1436), teacher of Qāzizāda Rūmī. This work, which reflects the first phrase of the Ottoman astronomical tradition in its treatment of planetary theory (*hayʾa*), follows the Maragha school in the main, especially Tūsī; but Muḥammad Ṣāḥ's approach here is atomistic, and he does not attempt to resolve the tension between mathematical and natural-philosophical principles as identified by Birūnī and Ṣirāzī.

165 *Miftāḥ al-sa'āda*, vol. 1, pp. 340–346. Taşköprüzāde neatly defines each of these applications as follows: "Know that the ability to effect occurrences by psychic (*naḥsānī*) force alone is [termed] magic (*sihr*). If this be by means of the harnessing of celestial entities (*falakiyyāt*), it is [termed] planetary invocation; if by means of conjuncting celestial

Perhaps responding to Ibn Ḥaldūn's attacks in particular, the Ottoman encyclopedist strongly defends the legitimacy of both magic and alchemy, at least when done properly.¹⁶⁶ Moreover, while he classifies all occult sciences exclusively as natural sciences, like Ibn al-Akfānī and others, Taşköprüzāde includes a subdiscipline of astrology, conjunctions (*'ilm al-qirānāt*), as branch of the mathematical science of astronomy (*'ilm al-hay'a*), together with astronomical tables and ephemerides (*'ilm al-zījāt wa-l-taqāwīm*) and the science of catastrophic events (*'ilm al-malāḥim*), the last depending in the main on judicial astrology.¹⁶⁷ Nor does he neglect lettrism; he simply reclassifies most of its subdisciplines, including Quran magic (*'ilm ḥawāṣṣ al-Qur'ān*), letter magic (*'ilm ḥawāṣṣ al-hurūf*),¹⁶⁸ divine-names magic (*'ilm al-taṣarruḥ bi-l-hurūf wa-l-asmā'*, *'ilm al-taṣrīf bi-l-ism al-a'zam*)¹⁶⁹ and letter divination (*'ilm al-ḡafr wa-l-ḡāmī'a*, aka the comprehensive prognosticon of 'Alī), as branches of the quranic science of tafsir.¹⁷⁰ At the same time, the science of magic squares (*'ilm a'dād al-wafq*), the basis of talismanic and letter magic, is naturally classified as a branch both of arithmetic (or geometry) and of lettrism.¹⁷¹ But most significantly, and

virtues with terrestrial, talismans; if by means of the harnessing of natural properties (*al-ḥawāṣṣ al-ṭabī'iyya*), the science of properties (*'ilm al-ḥawāṣṣ*) (if done through recitation (*qirā'a*)) or terrestrial magic (*nīranḡāt*) (if done through writing (*kitāba*)), or *ruqā'* (if done by other means). If it be by means of harnessing pure spirits (*al-arwāḥ al-sādiḡa*), it is [termed] spirit magic (*al-azā'im*); if by means of summoning these spirits to appear in the form of apparitions (*qawālib al-aṣbāḥ*), the science of summoning (*'ilm al-istiḡdār*), aka jinn magic (*tashīr al-jinn*). As for [obtaining] information about events elsewhere, whether past, present or future, it is [termed] soothsaying (*'ilm al-kihāna*). And just as man is capable of summoning incorporeal entities (*muḡarradāt*), so too can he hide [corporeal entities] from the senses, which is [termed] the science of concealment (*'ilm al-iḡfā'*); this includes the ability to hide things present from onlookers, which is [termed] ingenious trickery (*ḡiyāl*)."

166 *Miftāḥ al-sa'āda*, vol. 1, p. 317. In his defense of alchemy he relies on Faḥr al-Dīn Rāzī in the main.

167 He cites as exemplars of *'ilm al-malāḥim* the prediction of the massively destructive and transformative conquests of Nebuchadnezzar (Buḡtunaṣṣar) in the pre-Islamic period and Chinggis Ḥan, Hülegü and Temür in the Islamic (*Miftāḥ al-sa'āda*, vol. 1, p. 364).

168 He cites 'Abd al-Raḥmān al-Biṣṭāmī as this science's leading theorist (*Miftāḥ al-sa'āda*, vol. 2, p. 547).

169 This science is associated with al-Būnī and al-Biṣṭāmī in particular (*Miftāḥ al-sa'āda*, vol. 2, pp. 548f.).

170 *Miftāḥ al-sa'āda*, vol. 2, pp. 525–529, 547–550.

171 Taşköprüzāde openly says as much: "It is said that the equilibration of numbers (*i'tidāl al-a'dād*) has effulgent properties, this by reason of the spiritual essence (*rūḡāniyya*) of those numbers or letters. Astonishing effects and occult operations may be produced therefrom,

despite his disinclination to fully mathematicalize lettrism in the manner of his eastern colleagues, Taşköprüzâde imports the signature Ibn Turkian *taşkîk al-ḥarf* schema to structure the *Miftāḥ al-saʿāda* as a whole.¹⁷²

Taşköprüzâde's encyclopedia thus represents the most exhaustive Arabic classification of the sciences, including the occult sciences, produced in the 10th/16th century. Its seminal influence aside, however, fellow Ottoman scholars could and did quibble with its presentation of certain sciences, though without questioning its classificatory schema as a whole. Instances of such quibbling may be found, for example, in the eclectic and popular encyclopedic work *Netāyic ül-fünūn ve meḥāsin ül-mütūn* of Nev'ī Efendi (d. 1007/1599), a prominent poet, scholar and tutor to Ottoman princes, as well as a commentator on Ibn 'Arabī's *Fuṣūṣ al-ḥikam*.¹⁷³ Written in Ottoman Turkish rather than Arabic, the *Netāyic ül-fünūn* treats of 14 sciences (or clusters of related sciences) as follows:

History (*tārīḥ*); philosophy (*hikmet*); planetary theory (*hey'et*); theology (*kelām*); legal theory (*uṣūl-ı fıkḥ*); legal differences between the Hanafi and Shafi'i rites (*ḥilāf beyn ül-imāmeyn*); tafsir; sufism (*teşavvuf*); oneiro-mancy (*ta'bīr-i ḥ'wāb*); spells, incantations and [spiritual] medicine (*rūkā ve aḥsūn ve ṭbb[-ı rūḥānī]*); [physical] medicine (*ṭbb*); agriculture (*filāḥat*); astronomy-astrology (*nücūm*); divination and augury (*fāl ve zecr*).¹⁷⁴

on the condition that [astrologically] appropriate times and auspicious hours are chosen. There exist numerous wonderfully useful books on this science well known to its practitioners. This science is a branch of arithmetic insofar as it involves the calculation of numbers, and a branch of the [natural] science of active properties (*'ilm al-ḥawāṣṣ*) by reason of its effects and benefits. You will (God willing) hear of *'ilm al-wafq*, and likewise *'ilm al-ḥurūf wa-l-taksīr*, in the section on the isolated sura-initial letters (*muqatta'āt al-suwar*)" (*Miftāḥ*, vol. 1, p. 374). In this promised later section he then emphasizes the status of magic squares as an essentially lettrist discipline associated with figures like al-Būnī: "This science may be classified as a branch of arithmetic, given its focus on the ordering of numbers, or as a branch of geometry, given its focus on equilibrating those numbers or letters in tabular form (*al-ḡadāwīl al-wafqīyya*). But since it is equally possible to classify it as a type of letter magic (*ḥawāṣṣ al-ḥurūf*) in view of its use of letters [as well as numbers], I have mentioned it [here] under the rubric of lettrism (*'ilm al-ḥurūf*), which itself falls under the rubric of the active properties of the Qur'ān (*ḥawāṣṣ al-Qur'ān*)" (*Miftāḥ*, vol. 2, p. 548).

172 See Melvin-Koushki, "Of Islamic Grammatology."

173 My thanks to Ahmet Tunç Şen for bringing this work to my attention. For a summary see Schmidt, "The Occult Sciences," pp. 221–228.

174 A long final section (pp. 187–215 in the University of Michigan manuscript copy, Isl. Ms. 680, copied in 1068/1685; see <http://catalog.hathitrust.org/Record/006818660>) translates

Nev'ī's most notable disagreement with Taşköprüzâde, whom he cites by name in the preface (together with 'Abd al-Raḥmān al-Bisṭāmī and Ġazālī), is on the scientific and religious legitimacy of astrology itself, queen of the natural sciences for Ibn al-Akfānī. Both scholars clearly distinguish between *hay'a* and *nuḡūm* as formal disciplines;¹⁷⁵ but where Taşköprüzâde lionizes the latter as an essential component of natural philosophy and parent science to the full range of mantic disciplines, including geomancy, Nev'ī wholly dismisses it as being against reason and against Shari'a. Specifically, he posits *'ilīm-i nūcūm* as comprising three epistemological categories: mathematical (*ḥesābiyyāt*), presumptive (*ẓanniyyāt*)¹⁷⁶ and estimative or imaginary (*vehmiyyāt*). The first provides evidence for the oneness of God; the second, dealing, for instance, with the cycling of the seasons, is also religiously valid; but the third, representing judicial astrology proper, is both scientifically unreliable and religiously invalid.¹⁷⁷ That is to say, Nev'ī here seems to throw his lot in with the likes of Ibn Ḥaldūn, casting astrology beyond the pale even of natural philosophy. Strangely, however, he is not otherwise anti-occultist: his treatments of oneiromancy, spells and incantations,¹⁷⁸ quranic bibliomancy (*feth-i muṣḥaf*), lots and physiognomy are uniformly positive, even enthusiastic. At the same time, the Ottoman scholar is careful to distinguish between these sciences and Ġāhili-era mantic arts like augury, omens (*'āfiyet*), ornithomancy, soothsaying, belomancy (*ezlām*), scapulomancy and spasmatomancy, all of which, he asserts, are forbidden by the Shari'a.¹⁷⁹

Ġazālī's tale of Bashīr and Ṣādān, which features a moralistic discussion briefly touching on many disciplines. I did not have access to Ömer Tolgay's edition at the time of writing.

175 Nev'ī, like Taşköprüzâde, appears to have identified with the Mīrim Çelebi circle, naming his sources for *hay'a* as follows (Isl. Ms. 680, p. 88): Aṭīr al-Dīn Abharī (d. 663/1265), Ġābir b. Aflaḥ (fl. 6th/12th c.), Abū Rayḥān Bīrūnī, Naṣīr al-Dīn Ṭūsī, Maḥmūd Ġaḡmīnī (fl. 7th/13th c.), Faḫr Allāh 'Ubaydī (d. 751/1350), Kamāl al-Dīn Turkmānī (d. after 755/1354), Qāẓizāda Rūmī, 'Alī Qūṣṣī, Sinān Paşa (d. 891/1486), Mīrim Çelebi.

176 Schmidt gives "*ṭab'iyāt*" here, presumably a defective reading ("The Occult Sciences," p. 225).

177 Isl. Ms. 680, pp. 169 f.

178 *Ruqā* and *afsūn* are here explicitly defined as an occult science (*'ilīm-i meknūn ve dürr-i maḥzūn*) much pursued by ruling, scholarly and religious elites, and popularly termed 'spiritual medicine' (*ṭibb-ı rūḥānī*) (Isl. Ms. 680, p. 152).

179 Isl. Ms. 680, pp. 179–189. Cf. the *Riyāz al-qulūb*, a Persian compendium of eight sciences produced by one Ṣukr Allāh Šīrvānī Munaḡḡim (d. after 910/1505) for Bāyezīd II (r. 886–918/1481–1512) and surviving in unicum form, which treats of sufism, logic, planetary theory, astronomy-astrology (*nuḡūm*), calculation, physiognomy (*qiyāfa*), poetry (*shī'r*) and logographs (*mu'ammā*); most notably, Šīrvānī therein declares astrology to be the

Drawing simultaneously on Mamluk and Ilkhanid-Timurid intellectual contexts, then, the Ottoman Arabic (and Turkish) encyclopedic tradition as represented by Taşköprüzâde most prominently and scholars like Nevî secondarily may be said to be quite conservative in comparison to its Persian counterpart: it is persistently Avicennan—the occult sciences are here neither mathematicalized nor sanctified—and more juristically assertive—even astrology cannot be universally rehabilitated.¹⁸⁰ (A popular Ottoman treatise on Hanafî dogma, Birgili Mehmed Efendi's (d. 981/1573) *al-Tarîqa al-Muḥammadiyya*, flatly declares both astrology and magic illegal.¹⁸¹) Yet such scientific and religious conservatism did little to dampen interest in arts occult among Ottoman ruling and scholarly elites; indeed, some Iranian occultists found the Ottoman (or Mughal) court a far more attractive employer than the Safavid during the 10th/16th century.¹⁸² Given the paucity of comparative research on Ottoman, Safavid and Mughal intellectual history, of course, the significance of this Ottoman refusal to mathematicalize and sanctify the occult sciences—and the Ottoman *de-mathematization* of astronomy—cannot here be gauged; I can only speculate that it relates to the understandable Ottoman tendency to absorb the irresistible cultural influence of Iran, whether Timurid/Aqquyunlura or Safavid—, in somewhat reactionary fashion, as well as their adoption, through conquest, of Mamluk Arabic intellectual culture.¹⁸³ As that may be,

nooblest of all sciences, with the exception of theology. Such a statement closely tracks Bāyezîd's own celestial interests: he employed an unprecedented number of *munaḡğîms* at court and commissioned numerous astronomical-astrological texts and instruments (Şen, "Reading the Stars").

- 180 This unique sensitivity to astrology is emblemized by the demolition of the Istanbul Observatory in 987/1580: founded a few years prior due precisely to courtly enthusiasm for astrology, it was torn down to punish its director's incorrect prediction of an Ottoman defeat of the Safavids, which he argued was portended by the famous comet of 985/1577 (El-Rouayheb, *Islamic Intellectual History*, pp. 18 f.).
- 181 Schmidt, "The Occult Sciences," p. 221. Birgili's stern orthodoxy notwithstanding, however, it should be noted that his view of the rational sciences was highly nuanced, and his rejection of astrology was counterbalanced by his approval of astronomy (El-Rouayheb, *Islamic Intellectual History*, pp. 13–18).
- 182 A prime example here is Ḥaydar Rammāl, erstwhile tutor to Shah Ṭahmāsb, who decamped to Anatolia to become personal geomancer to Sultan Süleymān (r. 926–974/1520–1566) and there acquired immense wealth and influence (see Fleischer, "Seer to the Sultan").
- 183 That is, it is significant in this context that the mathematization of astronomy and the mathematicalization of the occult sciences were both projects launched in Iran. This is not to suggest that countervailing tendencies were not present there as well; as Fazlîoğlu

the fact remains: occultism stood as a central and unifying feature of High Persianate culture as a whole, from Lahore to Qazvin, from Isfahan to Istanbul, and was eagerly pursued by persophone, arabophone and turcophone elites.

Occultism and the History of Islamicate Philosophy-Science

While by no means exhaustive, this survey of the Persian encyclopedic tradition clearly shows that the occult sciences, and especially astrology, lettrism and geomancy, underwent a process of mathematicalization and sanctification, which process climaxed in Safavid Iran and Mughal India during the 10th/16th century. (Ottoman Anatolia, as noted, is here the odd man out.) To be sure, individual encyclopedists mathematicalize and/or sanctify various sciences according to taste, intellectual outlook or royal patronly needs, but the overall trend remains unmistakable and quite incontrovertible: over the course of the 7th–11th/13th–17th centuries, the progressive mathematicalization-sanctification of occultism marks its ascent to intellectually and politically mainstream status, and during this period the occult sciences only increased in prestige and scientific legitimacy throughout the Persianate east.

Yet such developments have been systematically ignored in the historiography of science to date, which is often at pains to paper over the increasingly heavy investment in occultism by the scholarly and ruling elites of the High Persianate period as massively reflected in the manuscript record. As a result, a large percentage of Islamicate intellectual production over many centuries—if the above encyclopedias are any indication, as much as one-sixth overall, and a considerably higher percentage of texts on the mathematical and natural sciences—is either quietly dismissed as irrelevant to any serious intellectual history or, more commonly, paraded as proof of “postclassical” Islamicate cultural decadence.¹⁸⁴ Manfred Ullmann clearly assumes, for instance, that the natural sciences as practiced in the premodern Islamicate world were so terminally infected by occultism as to render even “legitimate” sciences like medicine and astronomy crippled and finally abortive.¹⁸⁵ Fuat Sezgin, not content with mere positivism, attacks Ullmann for his sad acknowledgement

argues, the strong reassertion of Avicennism by the Daštakīs in Iran around the turn of the 10th/16th century “was a reaction to the mathematical humanism embraced in Samarqand and the kalāmīc natural philosophy followed in Istanbul,” particularly by Mīrim Çelebi (“Between Reality and Mentality,” p. 35).

184 Epitomizing the latter tactic is Abel, “La place.”

185 See e.g. *Natur- und Geheimwissenschaften*, p. 1.

of magic's ubiquity in Islamdom, counterposing a thoroughly whiggish triumphalism that excises the religio-intellectual cancer that is occultism by acknowledging only the achievements of valiant Muslim thinkers laboring to preserve "real" science—Greek, not eastern (Persian and Indian), and certainly not occult; thus only was Arabic science able to transmit the torch of the classical Greek heritage to Europe, subsiding into irrelevance after 430/1038.¹⁸⁶ Nor are Ullmann and Sezgin unrepresentative; even scholars like David Pingree, E.S. Kennedy and George Saliba, whose contributions to the history of Arabic science—including the history of astrology—are foundational and whose plumbing of the sources is more profound, often display the same tunnel vision that excludes certain bodies of data, especially occultist data, as irrelevant, even pesky.¹⁸⁷

While in most cases this defect does not impair the reliability of their findings, which are frequently groundbreaking (as with the crucial studies by Saliba and Ragep cited at the beginning of this article), it does result in tendentious, vertical narratives that float free of historical context,¹⁸⁸ as well as in whole volumes on the sciences in Islam that quietly censor the occult.¹⁸⁹ Perhaps the most unfortunate example of the latter is Ahmad Dallal's recent and otherwise useful synthetic overview of the current historiography of Islamicate science, which refers four times in merest passing to astrology, explicitly of worth only as a foil for explaining the rise of astronomy, and mentions ubiquitous sciences like alchemy and magic but once—as being forever exiled to the shadowlands of unscience by Ibn Ḥaldūn.¹⁹⁰ This argument alone, formally identical to the long-outmoded claim that Ḡazālī (d. 505/1111) was responsible for the permanent crippling of philosophy within Islamdom, stands eloquent testament to the fact that the state of the field with respect to occultism could not be sorer.¹⁹¹

186 See e.g. *Geschichte des arabischen Schrifttums*, vol. 7, p. 5 n. 2.

187 Lemay, "L'Islam historique," p. 20. Kennedy, for instance, habitually refers to astrology as a pseudoscience, worth studying only because it involves non-trivial mathematical skill, and wholly ignores closely related sciences like geomancy; see e.g. "The Astrological Houses," p. 535.

188 Brentjes and Morrison, "The Sciences," p. 564.

189 See e.g. the volume edited by J.P. Hogendijk and A.I. Sabra, *The Enterprise of Science*.

190 *Islam*, pp. 144 f. Amazingly, Dallal later dismisses modern forms of numerological analysis of the Quran—a mainstay of lettrism for over a millennium—as "cultic" (ibid., p. 172).

191 The fault for this lies not with Dallal, of course, who is simply summarizing the state of the field, relying on Saliba's findings in particular; and like Saliba, his intent is precisely to defend the honor of "postclassical" Islamicate science against the depredations of whiggish triumphalism.

At times, moreover, this tendentiousness strays from the territory of benign neglect into that of outright error. The most relevant example here is Saliba's contention in his seminal and summary article on the relationship of astrology to astronomy within the history of Islamicate science:

In the usual classifications of the sciences, the two disciplines [i.e., astronomy and astrology] were commonly considered as branches of the same science. It is only at a later date, around the thirteenth century, that astronomy was made fully distinct from astrology. Astronomy was thereafter counted as a mathematical science; and astrology was shifted to the applied physical sciences, together with agriculture, medicine, and alchemy.¹⁹²

As we have seen, precisely the opposite happened: occult sciences like astrology and geomancy, and even on occasion alchemy, were gradually shifted from the natural to the mathematical sciences in classifications of the sciences from the 6th/12th century onward, although only in the Persian tradition. (Saliba clearly considers only Arabic encyclopedias to belong to the category of, as he calls them, "the usual classifications of the sciences"—a highly problematic assumption, especially given his celebration of the astronomical tradition in Timurid-Safavid Iran; and in any event, astrology is consistently classified as a natural science in the Arabic encyclopedic tradition from Ibn Sīnā to Taṣḳöprüzāde.) Yet the above assertion is key to Saliba's larger argument that astronomy was progressively divorced from astrology, which was marginalized in all good society; thus freed of its religiously and scientifically suspect baggage, astronomy was then fully mathematized and hence "modernized" by a series of scientists in the Persianate world during the 7th–10th/13th–16th centuries.¹⁹³ Even a brief look into the contemporary Persian encyclopedic tradition, however, suggests that such a conclusion is premature, especially with respect to the 10th/16th century; future research into the question—research that does not isolate astrology from its closely associated disciplines of geomancy, letter divination and letter magic—will reveal, I suspect, a rather different picture. In the first place, the astronomy-astrology divorce hypothesis simply does not account for all the data, especially with respect to Iran and Transoxania. The state-of-the-art star tables produced there, for instance, including Tūsī's *Zīj-i Ṭlḥānī* and Ulūğ Beg's *Zīj-i ǧadīd-i sulṭānī*, comprise lengthy sections

192 "Astrology/Astronomy, Islamic," in idem, *A History of Arabic Astronomy*, p. 66.

193 See e.g. Saliba, "Islamic Astronomy."

with numerous tables devoted to astrological applications;¹⁹⁴ Ibn Turka himself reports, moreover, that the perennially popular science of astrology flourished in the Timurid period as never before as a direct result of the efforts of Qāzizāda Rūmī and Qūščī:

Astrology (*nuğūm*), whose basis is [astronomy (*‘ilm-i hay’at*)], ... is virtually the only topic of discussion in royal majlises, especially now that this science has undergone a renaissance (*iḥyā-yi ‘in ‘ilm az naw shuda*) in our own day; this is by virtue of the observatory founded [by Uluğ Beg, a feat] unmatched by sultans in centuries past.¹⁹⁵

Saliba’s motive in so arguing is an eminently worthy one, to be sure: to explode the whiggish claims of Sezgin and others that true science died in the Islamicate world after the 5th/11th century, counterposing this later period as the golden age of Islamicate science.¹⁹⁶ Indeed, my argument here is predicated on his powerful defense of “postclassical” Islamicate intellectual history. But such critiques of the decline thesis, however valiant in the face of its zombie-like persistence, do not escape or explode its basic teleology: Saliba still shares Sezgin’s sense of what “true science” must be—the more mathematical and less occult the better. That the occult is frequently mathematical by reason of intrinsic neopythagoreanism is not acknowledged. Given this shaky premise, Saliba’s lionization of such “early modern” Muslim heroes of science as Qāzizāda Rūmī, ‘Alī Qūščī and Šams al-Dīn Ḥafri, while necessary and most welcome, does them a disservice in one important respect: it bodily removes these thinkers from their respective Timurid and Safavid intellectual, social and political contexts, making it difficult to assess their motives and self-positioning within Islamicate intellectual history. *Why* did these thinkers mathematize astronomy? Were they primarily reacting to theological pressure or responding to royal interest, or was mathematization (and hence mathematicalization) simply the scientific thing to do for other reasons?

I argue that it is only by enlarging our aperture of analysis to include occultism and its sanctification that we can begin to address the highly complex—

194 Şen, “Reading the Stars.”

195 *Nafsat al-maṣdūr-i aval*, p. 211 (that astrology and astronomy are respectively meant is made clear by the passage’s context); see Melvin-Koushki, “The Quest,” p. 64. In other words, Ibn Turka considers astronomy and astrology to be a scientifically and sociopolitically cohesive unit, such that the renaissance of one must entail the renaissance of the other.

196 See e.g. Saliba, *Islamic Science*, pp. 233–255.

and ultimately unknowable—question of motive. That is to say, it is now time to investigate the horizontal axes of Islamicate intellectual history with the same vigor as has been devoted to the vertical.¹⁹⁷ Astronomer-philosophers like Qūščī and Ḥafīrī were living in a time and place of remarkable intellectual and cultural florescence, a significant portion of which was focused on sciences like lettrism, astrology and geomancy: they were operating intellectually and socially within a neopythagoreanizing and neoplatonizing zeitgeist. (Fazlhoğlu, for his part, argues for neoplatonism alone as philosophical basis for the mathematical humanism of the 9th/15th century, to the exclusion of the “mystical Hermetic-Pythagorean interpretation of numbers,” but I would suggest that such a clean distinction is unwarranted.¹⁹⁸) Far from being pulled forward along the arrow of progress by the siren song of incipient modernity, these ambitious thinkers were responding to and building on the intellectual resources they had to hand, as well as capitalizing on, or attempting to capitalize on, the interests and needs of royal patrons. We know fairly precisely what those immediate intellectual resources and political pressures were, moreover: together with the mathematizing astronomers Bīrūnī and Quṭb al-Dīn Šīrāzī as models, they were reading the works of neoplatonizing-neopythagoreanizing colleagues or teachers like Ibn Turka and Davānī, and enjoying the patronage of saint-philosopher-kings like Iskandar Sulṭān, Uluğ Beg, Ṭahmāsb, Humāyūn and Akbar, all of whom were profoundly attached to the occult sciences as a matter of millenarian imperial policy. Qāzīzāda Rūmī himself was remembered as a disciple of Aḥlāṭī until at least the early Safavid period.¹⁹⁹ And as for Ḥafīrī, his status as father of the most prolific Persian writer on lettrism of the 10/16th century cannot be considered irrelevant to his own scientific-philosophical project. Of course, much further research remains to be done before this new picture of High Persianate culture can be fully fleshed out and

197 My thanks to Daniel Stolz for this observation; see his forthcoming *The Lighthouse and the Observatory*.

198 “Between Reality and Mentality,” p. 33. Fazlhoğlu also speculates that the need to bring mathematics into conformity with religious requirements led to its purging of “Pythagorean and mystical elements” in order to develop “a new, formal, symbolic language that could be considered religiously neutral”—that is to say, wholly departing from the approach of the Iḥwān al-Šafā’ (ibid., p. 31). I here posit the contrary thesis: that it was precisely the burgeoning of neopythagorean sensibilities, simultaneously “mystical” and philosophical, that rendered mathematics religiously legitimate, though certainly not “religiously neutral”—witness the new status of lettrism in particular as *the* science of *walāya* throughout the Persianate east.

199 See n. 32 above.

its mechanics established, though current scholarship already points strongly in this direction.²⁰⁰ But my basic thesis here, to wit, that the mathematical turn in astronomy was part of the neopythagorean turn in philosophy-science more generally, rests on the assumption that intellectual currents are more likely to flow together than to flow apart, especially in a daring, expansive, millenarian, monist age in which the impulse to encompass and universalize defined all elite pursuits: empire, sainthood, occultism, philosophy, belles-lettres.²⁰¹ Given their profound interdependence, therefore, quarantining these categories from one another is unlikely to be productive.

There is a problem of chronology, however. It is currently thought that the neoplatonic-neopythagorean turn in philosophy began in Iran in the late 9th/15th century and did not hit its stride until the turn of the 11th/17th—that is, it seems to be a strictly high Safavid-era phenomenon.²⁰² This, naturally, is far too late to provide a natural context for the mathematical turn of the mid-9th/15th century, which suggests it to be an unrelated development. In an important survey article, Sabine Schmidtke and Reza Pourjavady provide the basic narrative as follows: Philosophical trends during the 10th/16th century essentially continue and extend the Davānī-Daštakī models (the so-called school of Shiraz); by the end of that century, however, the need for a distinctively Shiʿi philosophy became more pressing, and so Iranian scholars began to build a pointedly neoplatonic platform in order to disentangle themselves from the Sunni *kalām* tradition, heavily Avicennan in orientation. This strategy entailed the strong identification of Shiʿi doctrine with Greek, pre-Avicennan philosophy, which was often pursued along expressly religious lines: many

200 See n. 157 above.

201 The same period, of course, saw the emergence of the *ṭarz-i tāza* or ‘fresh style’ in Persian poetry, long dismissed as decadently ornate and overcomplex through infection by Indian influences—hence its epithet ‘the Indian style’ (*sabk-i hindī*). Further marginalized in the wake of the 13th/19th-century neoclassical (*bāzgašt*) movement in Iran, it is currently being rehabilitated by literary historians like Paul Losensky and others. A similar project is under way on the ornate and encyclopedic Mamluk Arabic literary tradition. It is highly relevant in this connection that the leading lettrists of the 9th/15th century were also acclaimed as the greatest ornate prose (*naṣr-i fannī*) stylists of their generation, including ‘Abd al-Raḥmān al-Bisṭāmī in Arabic and Ibn Turka and Šaraf al-Dīn Yazdī in Persian—suggesting that occultist logocentrism informed, and in some cases drove, the ornate literary (and calligraphic) culture of the era (Melvin-Koushki, “The Quest,” pp. 380–388).

202 And a late Mughal-era phenomenon by extension: Mīr Dāmād and Mullā Šadrā represent the twin philosophical schools that became mainstream in India by the 12th/18th century, with the direct influence of Davānī and the Daštakīs laying the necessary groundwork in the 10th/16th (Rizvi, “Mīr Dāmād,” p. 11; Pourjavady, *Philosophy*, pp. 12 f.).

important Safavid philosophers were also high-ranking and influential religious authorities, and dedicated to the comprehensive exposition of the new and expanding Twelver Shi'i hadith canon in particular. The growing interest in Graeco-Arabica during the early Safavid period thus led to the identification of the teachings of the Imams with those of the Greek ancients, especially Pythagoras, Plato, (Ps.-)Aristotle and Alexander of Aphrodisias. This penchant for excavating earlier layers of the philosophical tradition, and especially the "pure" philosophy of the pre-Socratics, had of course been characteristic of the illuminationist strand of philosophy from the 7th/13th century onward; but it is during the Safavid period that the Graeco-Arabica business boomed, with (Ps.-)Aristotelica being particularly important—including in the first place the *Uṭūlūḡiyyā*. Davānī was the first to bring the *Theology* back into mainstream philosophical use in Iran (he cites it repeatedly, for example, in his first *ḥāšīya* on Qūščī's commentary on Ṭūsī's *Taḡrīd al-i'tiqād*); Ġiyās al-Dīn Daštakī then produced an abridgment of and glosses on the text. The *Theology* increased in popularity with their students and their students' students, finally reaching peak influence by the early to mid-11th/17th century, as indicated by a proliferation of copies produced then. We should therefore speak of an *antiquarian turn* as guiding the development of Safavid philosophy—a similar turn, intriguingly enough, informing contemporary trends in European Renaissance philosophy, which was also neoplatonizing (not to mention heavily occultist and kabbalistic).²⁰³

But does this thoroughgoing neoplatonization of philosophy in Iran really only begin with Davānī? Is his antiquarianism simply of the standard illuminationist type? And is the Safavid-era positing of the religio-philosophical identity of the Imams and the ancients really unprecedented? It seems not: Davānī declares his dependence on Ibn Turka in particular, for it is Ibn Turka and his imamophilic lettrist colleagues who, in their quest to systematize sanctified occultism, powerfully assert the doctrine of *prisca theologia* whereby the Imams and the ancients stand equal as sources of all occult knowledge. It so happens, moreover, that in Ibn Turka's reading this *philosophia perennis* is quintessentially neopythagorean-neoplatonic. While these new Brethren of Purity of the early 9th/15th century were not engaged in textually excavating Greek philosophy in the way that became standard in the later Safavid period, then, and while they were most certainly not Shi'i,²⁰⁴ the lettrist revolution

203 Pourjavady and Schmidtke, "An Eastern Renaissance?" pp. 254–259.

204 *Imamophilia*, termed Alid loyalism by Marshall Hodgson and Twelver Sunnism (*tasannun-i isnā-'ašarī*) by Muḥammad-Taqī Dānišpažūh and Rasūl Ġa'fariyān, is a distinctive post-Mongol, pre-Safavid phenomenon that, significantly, was coeval with the occultist renaiss-

they spearheaded helped to shape the intellectual terrain of Timurid-Turkmen Iran and continued to reverberate for centuries thereafter. (The great Qajar-era reviver of Sadrian philosophy, Mullā ‘Alī Nūrī (d. 1246/1831), for instance, thoroughly glossed Ibn Turka’s *Book of Inquiries*, suggesting its status as a canonical text for centuries.²⁰⁵) That is to say, this brand of sanctified neopythagoreanizing occultism, fundamentally antiquarian and imamophilic in ethos, was the only 9th/15th-century Iranian Sunni intellectual current immediately compatible with Twelver Shi‘ism, and so was carried over by Safavid philosophers virtually wholesale; the roots of Safavid philosophy therefore lie partially in late Mamluk Cairo and its occultist renaissance.

While instances of this intellectual continuity could be multiplied, it is particularly explicit in the oeuvre of Mīr Dāmād. Consciously embracing the Ibn Turkian legacy, Mīr Dāmād wrote three works wholly or substantially lettrist in approach, including in the first place the *Ġazavāt u mavāqīt*, his only major work in Persian, which offers a full, and characteristically ornate, treatment of his philosophical system within a neopythagorean-lettrist framework.²⁰⁶ Here as elsewhere, Mīr Dāmād compares and contrasts what he calls his Yemeni philosophy (*al-ḥikma al-yamāniyya*)²⁰⁷ with Greek precedent (*al-ḥikma al-yūnāniyya*), here represented in the first place by Pythagoras and the neopythagoreans (called by him *fīṣāḡūrasīyyīn* or *āl-i Fīṣāḡūras*), together with Aristotle (i.e., Plotinus), Plato, Nicomachus, Archytas of Tarentum, Anaximenes, Ptolemy, Homer and the Stoics. His primary sources include Ibn Sīnā’s possibly pseudepigraphal and essentially neopythagorean *R. al-Nayrūziyya*, Ibn Turka’s *R. Ḥurūf* and Davānī’s *Tuḥfa-yi rūḥānī fī ḥavāṣṣ al-ḥurūf*, a manual of astrological letter magic which itself cites Ibn Turka as a source, and he freely and accurately quotes from all three. Not coincidentally, it is in the opening section of the *Ġazavāt* that Mīr Dāmād declares the *Uṭlūḡiyyā* to be fundamental to his entire philosophical system.²⁰⁸ This summary declaration is followed by others, including this epitome of pure neopythagoreanism:

sance (see Melvin-Koushki, “The Quest,” pp. 69–77). The emergence of a new Sunni-Shi‘i binary during the Ottoman-Safavid conflict notwithstanding, moreover, this phenomenon seems to have experienced a revival in Mughal India during the 11th–12th/17th–18th centuries (my thanks to Mana Kia for this observation).

205 Melvin-Koushki, “The Quest,” pp. 437, 573 f.

206 His other two lettrist works, both in Arabic, are *Nibrās al-dīyā’* (ed. Ḥāmid Nāḡī Iṣfahānī) and *Ta’wīl al-muqaṭṭa’āt*.

207 His source for this term is the Prophetic hadith *Faith is Yemeni, and wisdom is Yemeni* (*al-īmān yamānī^m wa-l-ḥikma yamāniyya*) (see e.g. Buḥārī, *Ṣaḥīḥ*, *Kitāb al-Maḡāzī* (46), *Bāb Qudūm al-Aṣ‘ariyyīn wa-ahl al-Yaman* (76), no. 4431).

208 *Ġazavāt u mavāqīt*, p. 15.

In sum: The realm of the letter corresponds to that of number, and the realm of number corresponds to that of engendered existence (*kawn*); and the relationships (*munāsabāt*) that obtain in the realm of letter correspond to those that obtain in the realm of number, which in turn correspond to the commixtures (*mumāzağāt, muḥālaṭāt*) that constitute the realm of engendered existence.²⁰⁹

Significantly for our purposes here, Mīr Dāmād further quotes Šāms al-Dīn Ḥafri's *Tafsīr Āyat al-kursī* for its specifically neopythagorean content: there the astronomer too invokes the ancient association between letters, numbers and the various orders of existence, which he attributes to Indian thinkers in particular.²¹⁰ From the perspective of the early 11th/17th century, then, the latter-day tradition of philosophical lettrism begins with Ibn Turka, who associates it with the Imams and the ancients as well as with Ibn 'Arabī in particular; is continued by transitional thinkers like Davānī and Ḥafri; then is brought to full polish by Mīr Dāmād, his students and his colleagues, Twelver neopythagoreans to a man.

Mīr Dāmād, it hardly needs saying, was not exceptional in his lettrist or antiquarian proclivities; that they pervaded Safavid intellectual culture as a whole may also be seen in the many philosophical anthologies that were produced in the 11th/17th century. One impressive anthology in particular epitomizes that culture: MS Dānišgāh, Dāniškada-yi Ilāhiyyāt, 242B. Containing 87 texts or text excerpts, its strong neoplatonic-neopythagorean flavor is produced by the juxtaposition of ancient philosophers like Ps.-Aristotle (including the *Uṭlūğīyā*), Alexander of Aphrodisias and Ps.-Apollonius with Muslim philosophers like al-Kindī, (Ps.-)Fārābī (including the *Ġam' bayn ra'yay al-ḥakīmāyn*), Ibn Sīnā (including the *R. al-Nayrūziyya*), Suhrawardī, Ṭūsī, Quṭb al-Dīn Širāzī, Davānī and Ḥafri, and an occultist spicing is further conferred by a number of alchemical texts from the Ġābirian corpus and, naturally, a work attributed to Ġa'far al-Šādiq on *ğafṛ*, the last in the anthology. That is to say, this compendium constructs an intellectual continuum in which occult sciences (here lettrism and alchemy) are considered essential components of philosophical practice, especially in a Shi'izing-neoplatonizing context.²¹¹

Nor were the occultist proclivities of prominent Safavid philosophers purely academic: Ġiyās al-Dīn Daštakī, Šayḥ Bahā'ī, Mīr Dāmād and others figure in

209 *Ğazavāt u mavāqīt*, pp. 133f.

210 *Ğazavāt u mavāqīt*, pp. 167–169; Ḥafri's tafsir has been edited by Ġavād Fāzil Baḥšāyiši.

211 Endress, "Ein-Band-Bibliotheken," pp. 35–56. Cf. the similar pairing of alchemy and kabbalah in contemporary Europe (Forshaw, "*Cabala Chymica*").

contemporary and later sources as powerful occultists in service to the court, and masters of letter or talismanic magic in particular.²¹² This point is of particular importance for understanding the nature of high Safavid imperial culture, which was largely constructed—and sanctified—by these philosopher-mages along the lines indicated by the encyclopedist-biographer-judge Abū l-Qāsim Kāzīrūnī: simultaneously Twelver Shi'i, sufi, occultist and neopythagorean-neoplatonic. The fact remains, however, that both theoretical and applied occultism in the Safavid-Mughal period was consciously derived by its practitioners from Timurid-Aqquyunlu precedent, itself an outgrowth of developments centering on late Mamluk Cairo, where occultism was first sanctified by sufi lettrists.

Conclusion

Whatever marginalization the occult sciences suffered under the brunt of attacks by philosophers like Ibn Sīnā, theologians like Ġazālī, traditionalists like Ibn Taymiyya and historians like Ibn Ḥaldūn, it was more than reversed in the post-Mongol period. From the mid-8th/14th century onward, occultism was increasingly a mainstream pursuit of scholarly and ruling elites throughout the Islamicate heartlands, due in large part to its sanctification through association with sufism, newly hegemonic, with Ibn 'Arabī and al-Būnī being pivotal, emblematic figures in this process. The occultist renaissance also drove—or was driven by—a general shift away from Avicennan aristotelianism toward a neoplatonism heavily neopythagorean in orientation; here we should take seriously Ibn Qayyim al-Ġawziyya's anxious characterization of a specifically neopythagorean-neoplatonic (*fiṭāḡūrī-aflāṭūnī*) brand of high occultism, together with Avicennan philosophy, as the greatest threat to Islam in the post-Mongol period—a polemical argument that nevertheless predicts High Persianate intellectual developments quite accurately.

212 See Melvin-Koushki, *The Occult Science*. The Safavid-era recension of Ḥusayn Vā'iz Kāshif's *Asrār-i Qāsimī*, for instance, was considerably expanded to include tales of these and other philosophers' occultist feats as performed for—or against—various clients among the ruling elite; see the 1883 Bombay lithograph, pp. 85, 87, 92, 97, 98, 99, 102, 104, 105. On this recension see Subtelny, "Sufism" (my thanks to Professor Subtelny for sharing a working draft of this article); Melvin-Koushki, *The Occult Science*. And it appears this memory has not faded with time: a number of Persian grimoires still circulate in cheap bazaar editions under the title *Mīr Dāmād-i kabīr*.

These trends are, not surprisingly, mirrored in Arabic and Persian classifications of the sciences produced between the 6th–11th/12th–17th centuries. While the Arabic encyclopedic tradition (primarily Mamluk and Ottoman) registers the florescence of occultism at the heart of elite Islamicate culture, it appears to be more scientifically and religiously conservative: there the occult sciences retain their standard Avicennan classification as derivative natural sciences, including disciplines like astrology, which remained divorced from the strictly mathematical science of astronomy; and juristic doubts about the licitness of certain occult sciences persisted in the face of overwhelming elite and popular interest. It is only in the Persian encyclopedic tradition, beginning with Faḥr al-Dīn Rāzī's *Ġāmi' al-ʿulūm*, that the occult sciences are progressively (if unevenly) mathematicalized and sanctified, such that by the 10th/16th century sciences like astrology, lettrism and geomancy are classified as both mathematical sciences and prime vectors of *walāya*. This development may be described as a resurgence of the expressly neopythagorean-occultist project of the 4th/10th-century Iḥwān al-Šafā', given new impetus through the sanctification of the occult sciences, especially lettrism, which in its reformulation in the early 9th/15th century by thinkers like Ibn Turka and 'Abd al-Raḥmān al-Biṣṭāmī—self-identifying as new Brethren of Purity—comes to epitomize the neopythagorean turn in its elevation to the status of universal science: number-letter as the root of all knowledge and all-creative matrix of the cosmos.²¹³

Not coincidentally, it was at precisely the same time that astronomers working at the Samarkand Observatory, including in the first place Qāẓizāda Rūmī and 'Alī Qūšċī, initiated what is hailed in current historiography of science as a mathematical turn that for the first time wholly freed astronomy from the constraints of aristotelian physics; this allowed for unprecedentedly trenchant critiques of the Ptolemaic geocentric model that may have inspired Copernicus. Current scholarship presents this mathematical turn as the outcome of attempts to sidestep theological critiques—a valid explanation, to be sure, but an insufficient one; it ignores changes in the broader intellectual and religiopolitical context as encapsulated in contemporary Persian classifications of the sciences, and particularly the large-scale trend toward the simultaneous mathematicalization and sanctification of the occult sciences. As a consequence, it smacks of the Victorian-era vivisectionist conviction that the categories *science*, *magic* and *religion* can be surgically separated and evolutionarily ranked.

213 The concurrent development of the Ġābirian-Ġildakian alchemical tradition was driven by a similar universalizing impulse, and intersected with lettrism in ways that have yet to be explored. Here a study of Aḥlāṭī—Mamluk lettrist-chemist *nonpareil*—remains a major desideratum.

It is this ingrained atomism, this religious frigidity,²¹⁴ that continues to drive scholarly occultophobia, rendering occultism the largest blind spot in the historiography of Islamicate science. If the argument holds, therefore, that an occult science like astrology was originally naturalized in order to dissociate mathematical astronomy from those parts of the Greek philosophical heritage thought to be religiously objectionable, then its mathematicalization and sanctification from the 6th/12th century onward signals its accession in some circles to not just scientific but also religious legitimacy: a crucial scientific tool in the spiritual quest, roundly pursued by ruling, scholarly and religious elites the Persianate world over, for *self-divinization*, *theosis* (*ta'alluh*). Nor were our astronomers lone geniuses working in a vacuum: they were colleagues to and correspondents with ardent neopythagoreanizers like Ibn Turka, which cannot but have influenced their intellectual outlook.

Far from faltering, moreover, the neopythagorean turn only increased in vigor in Iran during the Safavid period. It is well-known that Safavid-era philosophy is characterized by a florescence of neoplatonism and neopythagoreanism, this largely driven by the need to create a new Shi'ism-compatible philosophy distinct from mainstream Sunni Avicennism. This in turn drove what may be called an antiquarian turn among Safavid scholarly elites, who, taking a page from earlier illuminationists, increasingly sought to undercut the competition by turning directly to the sages of Greece, now closely identified with the Imams (*à la* Ibn Turka); the *Theology of Aristotle* and other neoplatonic Ps.-Aristotelica exploded in popularity, and Ibn Sīnā himself was neopythagoreanized. What has not yet been recognized, however, is the role of occultism generally and lettrism specifically in aiding and abetting this process. I suggest that the brand of imamophilic neopythagoreanizing occultism developed in 9th/15th-century Timurid-Aqquyunlu circles was the Sunni intellectual current most immediately utilizable by Twelver Shi'i scholars seeking to construct a new Safavid imperial culture—hence the lettrist demonstrations of philosophers from Davānī to Mīr Dāmād, and the acclaim in the sources of Ġiyās al-Dīn Daštakī, Šayḥ Bahā'ī and their peers as the greatest letter mages of the century. The sanctification of Iran as a sufi-Shi'i state is therefore not simply an upshot of the transformative Chinggisid era and its complex religious politics, but has roots even in 7th/13th-century North Africa, where lettrism became *the* science of *walāya* and hence core of the Imamic legacy for Shi'i and Sunni alike.²¹⁵

214 Kripal, *Authors*, p. 17.

215 On possible North African Isma'ili precedents see Ebstein, *Mysticism*; Gardiner, "Esotericism," ch. 3. Suggestive here is Ibn al-Qayyim's assertion that neopythagorean-neoplatonic

It is my contention in this article that the above developments are intimately related: I propose the occultist, and especially lettrist, renaissance of the turn of the 9th/15th century as the missing link that connects the mathematical turn in astronomy in the 9th/15th century with the neopythagorean and antiquarian turns in metaphysics in the 10th–11th/16th–17th, while also restoring High Persianate intellectual developments to their proper religiopolitical context. Each of these trends may be traced in large part to early Timurid Iran, and indeed earlier to late Mamluk Egypt; they then come to fullest flower in high imperial Safavid Iran, as well as, in slightly different guise, in high imperial Mughal India: continuity, not rupture.²¹⁶ The organic evolution of these intellectual currents is moreover trackable in the Persian encyclopedic tradition, one of the most fruitful areas for future research into “postclassical” Islamicate intellectual and cultural history.²¹⁷ As this and other bodies of evidence suggest, sanctified neopythagorean-neoplatonic occultism is part of the basic ligamentation of Islamicate philosophy-science and political practice during the High Persianate era; any historiography that severs it must hobble.

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occultism is the direct heir of Isma‘ilism in the post-Mongol period; as Livingstone puts it, the Hanbali theologian considered occultism to be a form of “Ismā‘īlī Pythagorean gnosticism,” the “superstitious offspring of legitimate science and Ismā‘īlī spiritual alchemy” (“Science and the Occult,” p. 608; see Melvin-Koushki, “Defending Geomancy”). Cf. al-Maqrīzī’s almost admiring references to Ḥasan-i Šabbāḥ as master occultist (see n. 5 above). On the development of Islamicate lettrism from its originary 1st/7th-century Shi‘i context onward see Melvin-Koushki, “The Quest,” pp. 171–283.

²¹⁶ This is not to imply a decline of lettrist thought in the Persianate world after the 11th/17th century; it continued to be pursued in Iran, for instance, by scholars like Mullā ‘Alī Nūrī (d. 1247/1832), the great Qajar-era reviver of Sadrian philosophy, and the Shaykhi, Babi and Baha’i movements were all profoundly lettrist in their respective cosmological outlooks (Melvin-Koushki, “The Quest,” pp. 281, 437). Lettrism specifically and occultism generally also remained a mainstream scholarly pursuit in Persianate Central Asia until the region’s annexation by the Soviet Union in 1920 (Melvin-Koushki and Pickett, “Mobilizing Magic”).

²¹⁷ Vesel’s otherwise excellent study of the Persian encyclopedic tradition unfortunately stops in the mid-8th/14th century with Āmulī’s *Nafāyis al-funūn*.

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